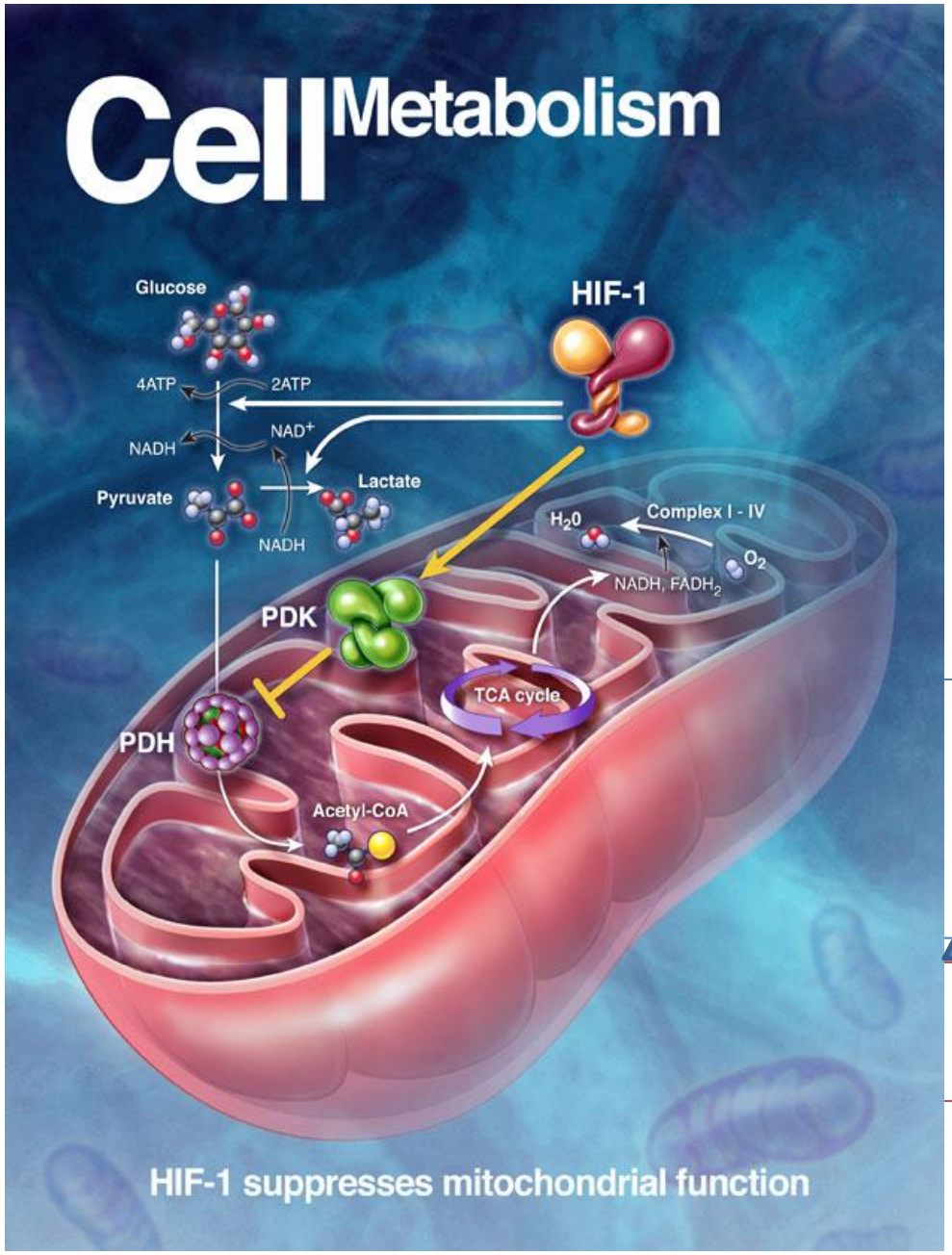


INTRODUCTION TO METABOLISM

Thousands of reactions are occurring simultaneously in a single cell.

DR. HUMIRA AMAN ALI

OBJECTIVES

- At the end of the lecture, student would be able to:
- Define metabolism and intermediary metabolism.
- Describes anabolism and catabolism.
- Briefly describe cellular respiration mechanisms.
- E:g Electron Transport Chain

Metabolism

- **Metabolism (from Greek: "metabolē", "change" or Greek: *metabolismos*, "out throw") is the set of life-sustaining chemical transformations within the cells of living organisms.**
- **These enzyme-catalyzed reactions allow organisms to grow and reproduce, maintain their structures, and respond to their environments.**

- The word metabolism can also refer to all chemical reactions that occur in living organisms, including digestion and the transport of substances into and between different cells, in which case the set of reactions within the cells is called **intermediary metabolism** or **intermediate metabolism**.

INTERMEDIARY METABOLISM

- **Definition of INTERMEDIARY METABOLISM:** the intracellular process by which nutritive material is converted into cellular components—called also *intermediate metabolism*

****

Metabolism is usually divided into two categories:

Catabolism **= the *breakdown* of complex substances.**

Anabolism **= the *synthesis* of complex substances from simpler ones or uses energy to construct components of cells such as proteins and nucleic acids.**

****

So generally catabolic processes generate energy for anabolic processes

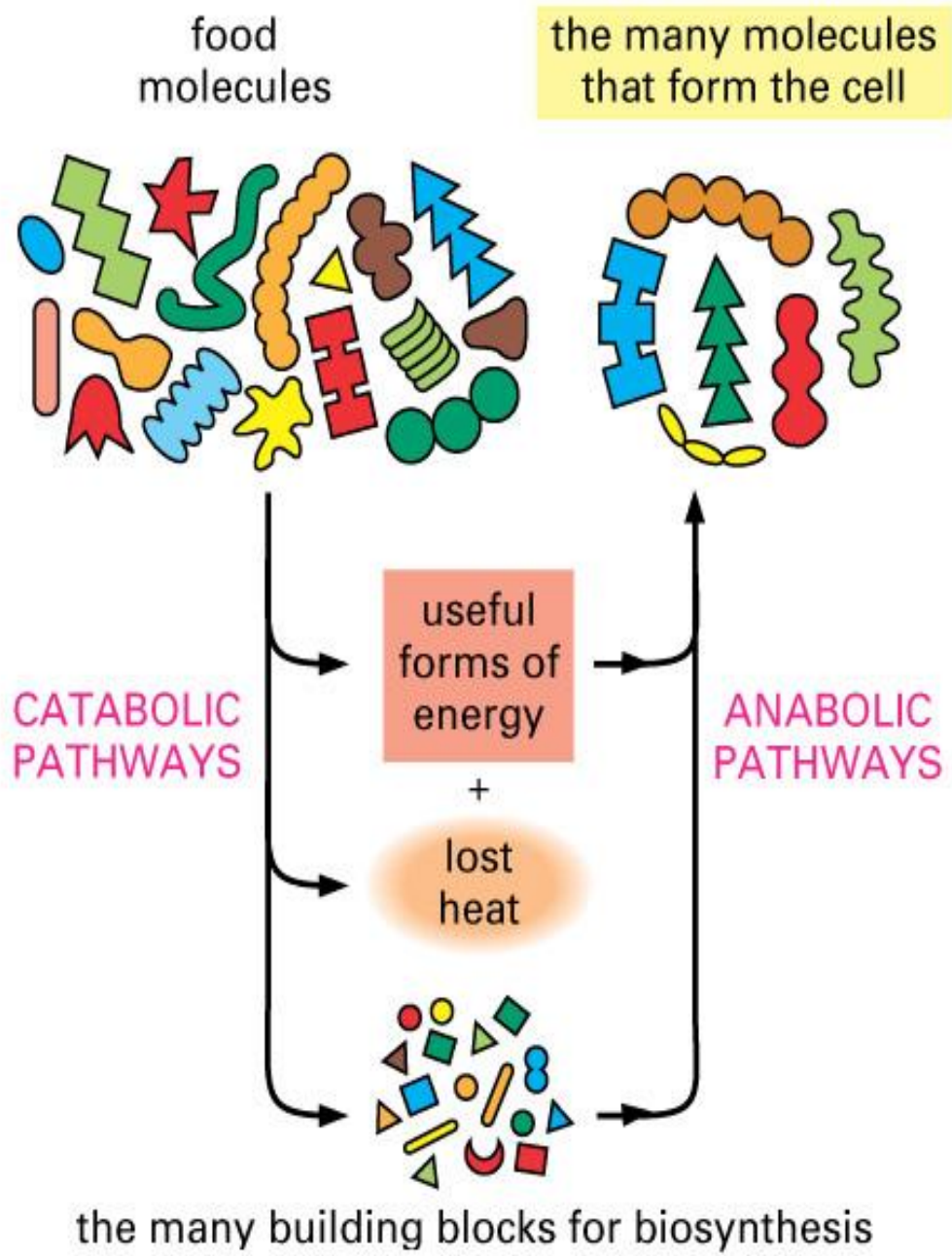
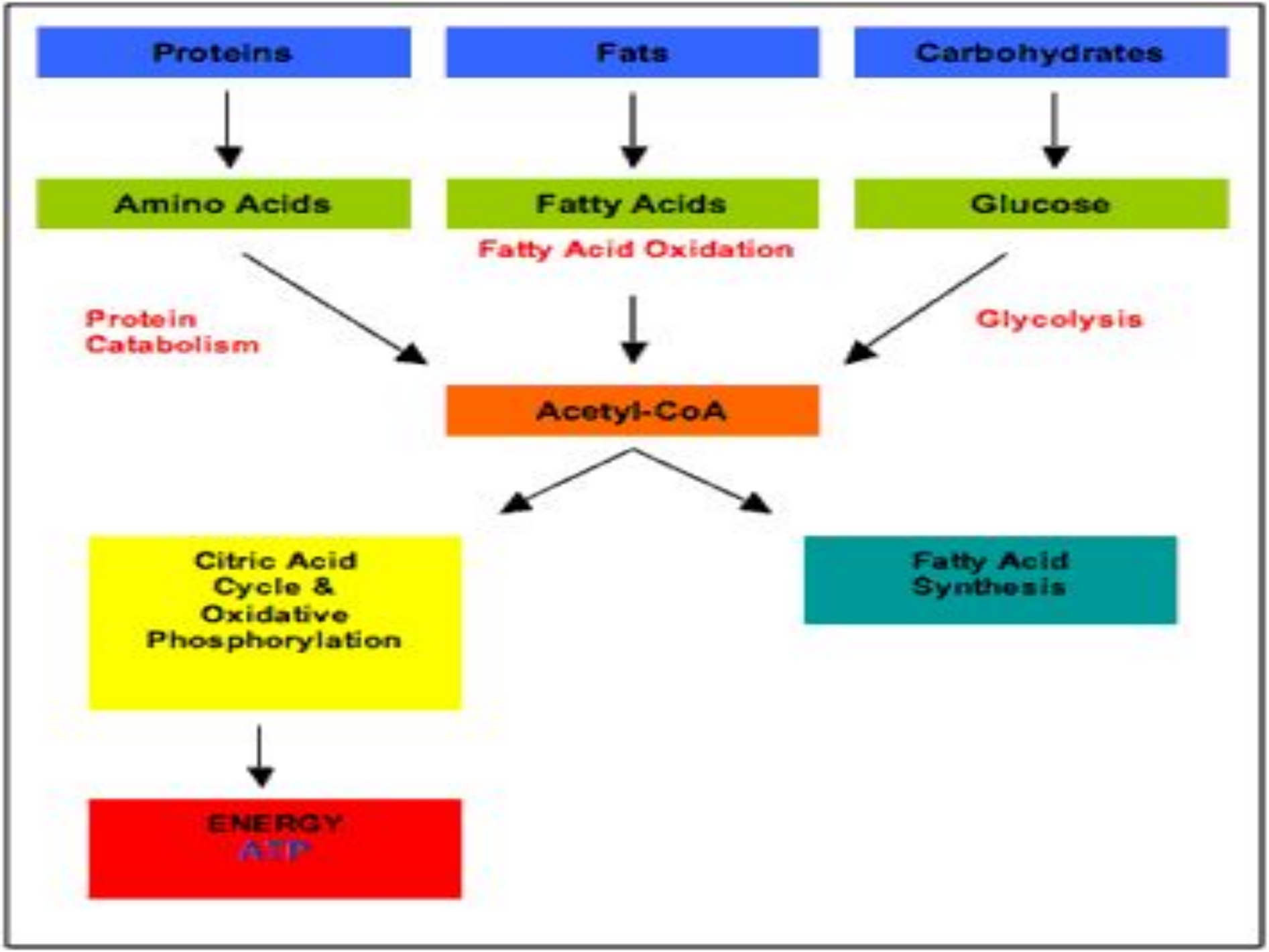


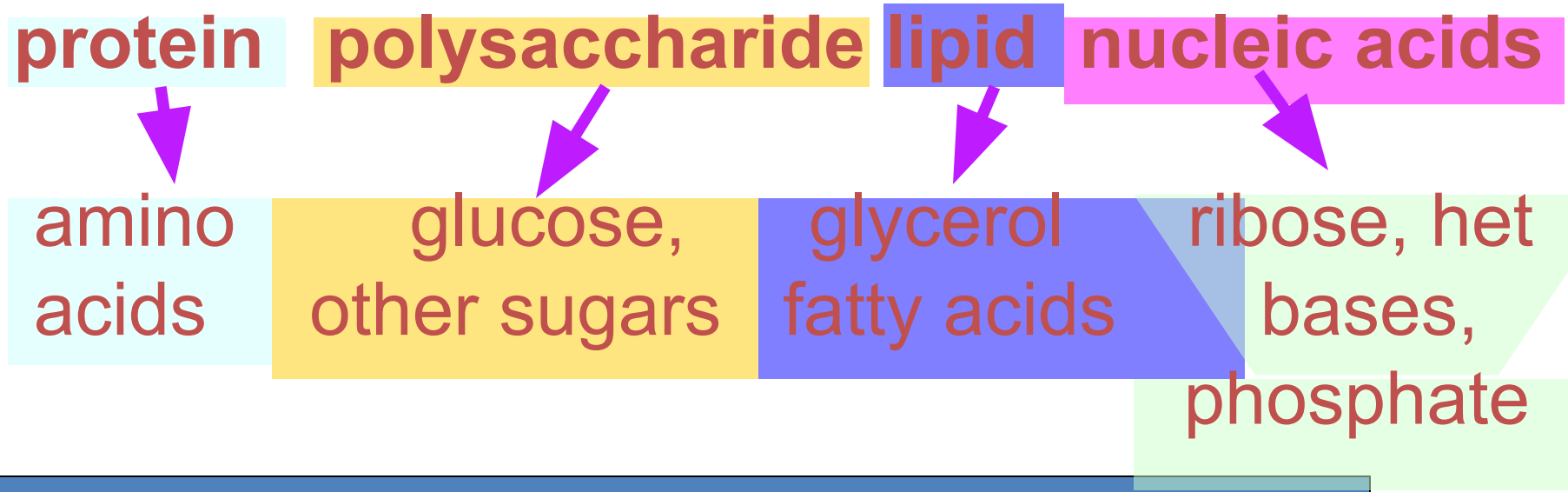
Figure 3-3 Essential Cell Biology, 2/e. (© 2004 Garland Science)


Proteins
Fats
Carbohydrates
Amino Acids
Fatty Acids
Glucose
Fatty Acid Oxidation
Protein Catabolism
Glycolysis
Acetyl-CoA
Citric Acid Cycle & Oxidative Phosphorylation
Fatty Acid Synthesis
ENERGY
ATP

General Pathways of Metabolism

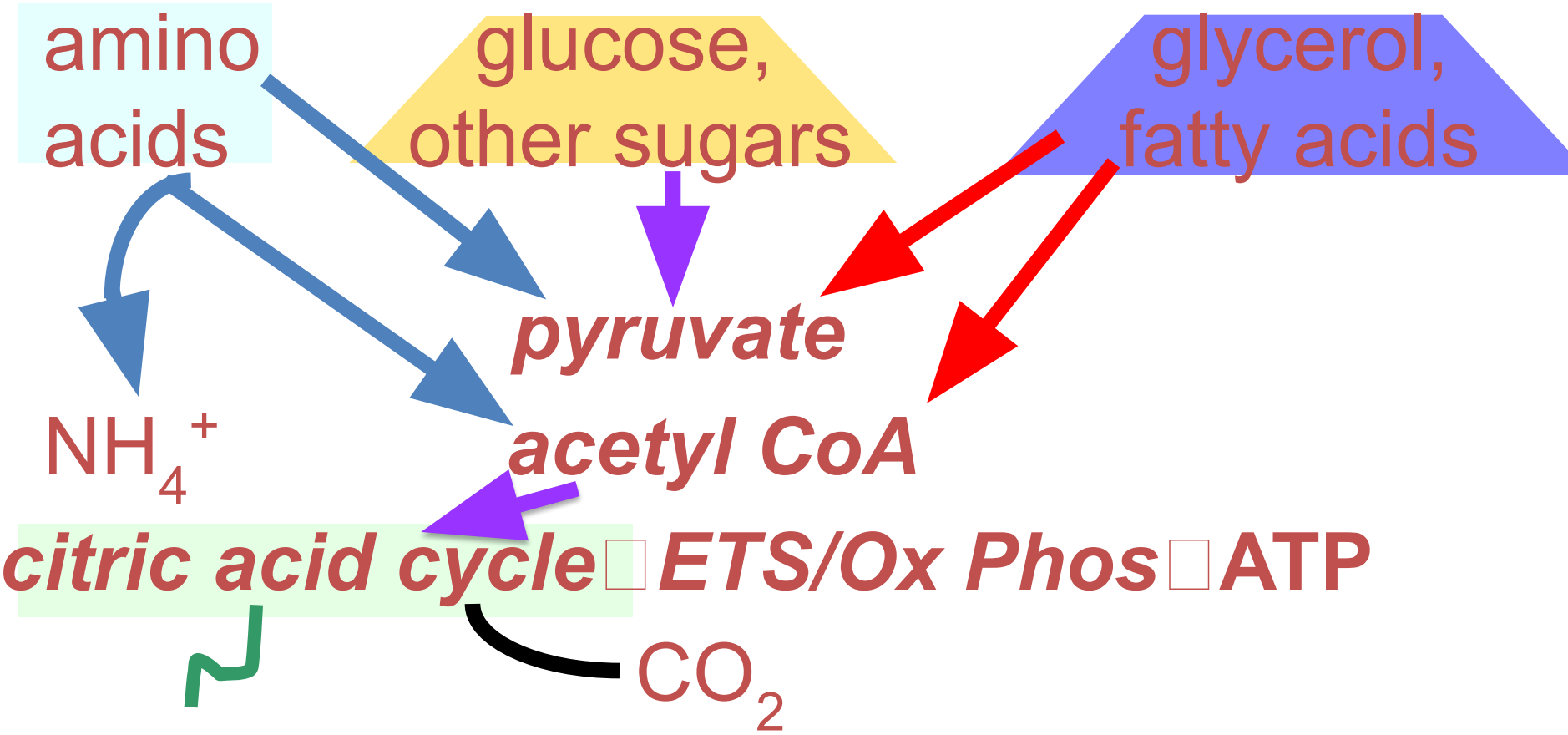
-- Catabolism --

1- Breakdown of macromolecules to building blocks -- generally hydrolytic



-- no useable energy yield here- only building blocks obtained

2- Breakdown of monomers to ***common intermediates***

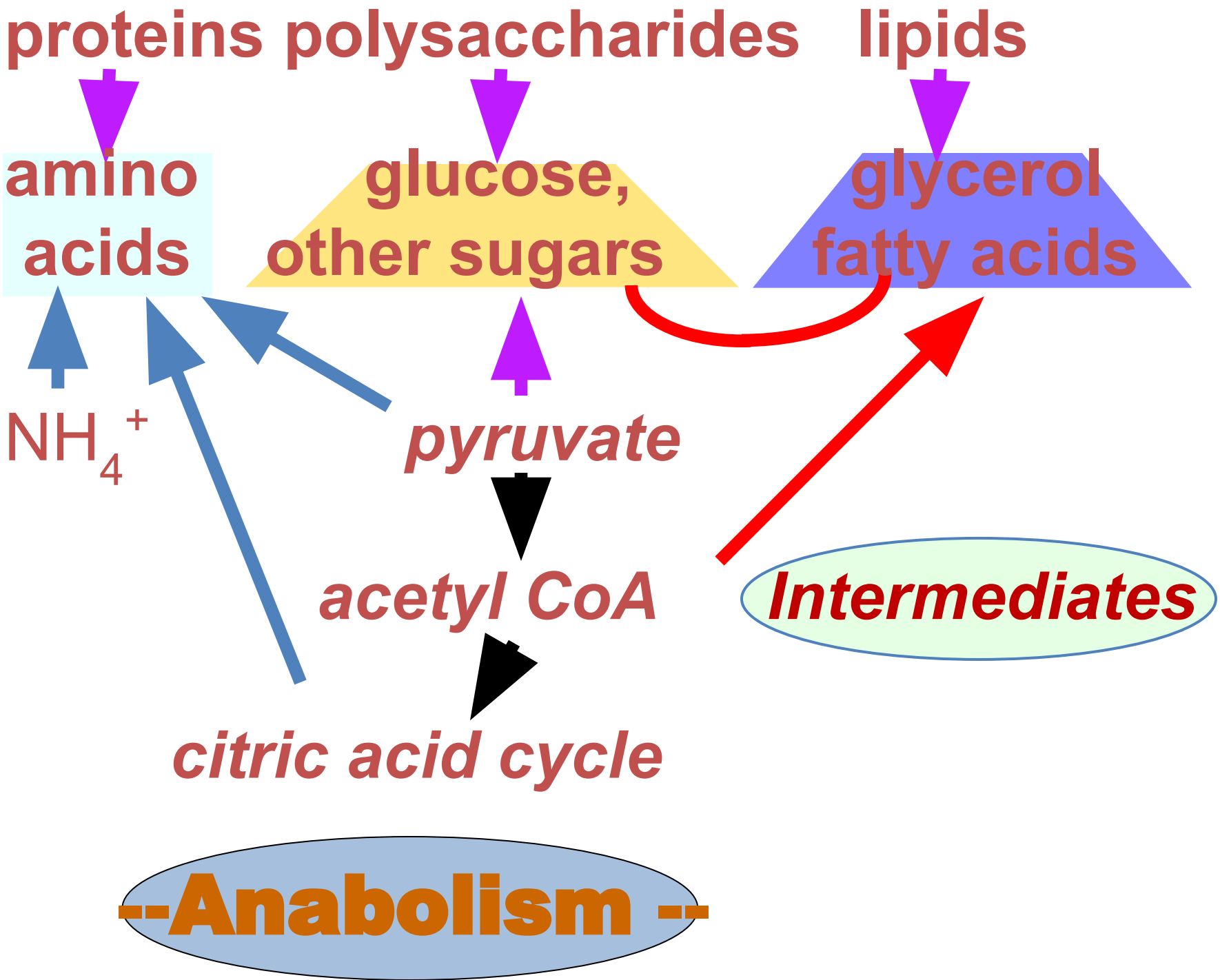


Oxidative processes -- produce **ATP & NADH for energy**

3-Breakdown of intermediates to CO_2 and electrons is accomplished through a central oxidative pathway:

- **the *Citric Acid Cycle* or *TCA* or the *Krebs Cycle*.**

- **This cycle leads to the production of ATP by processes called electron transport and oxidative phosphorylation.**


proteins
polysaccharides
lipids
amino acids
glucose, other sugars
glycerol fatty acids
NH_4^+
pyruvate
acetyl CoA
Intermediates
citric acid cycle
--Anabolism --

Anabolism

1- utilization of critical Common Intermediates including components of TCA cycle to make building blocks

2 - making building block requires energy = ATP

3 - synthesis of macromolecules requires energy = ATP

****

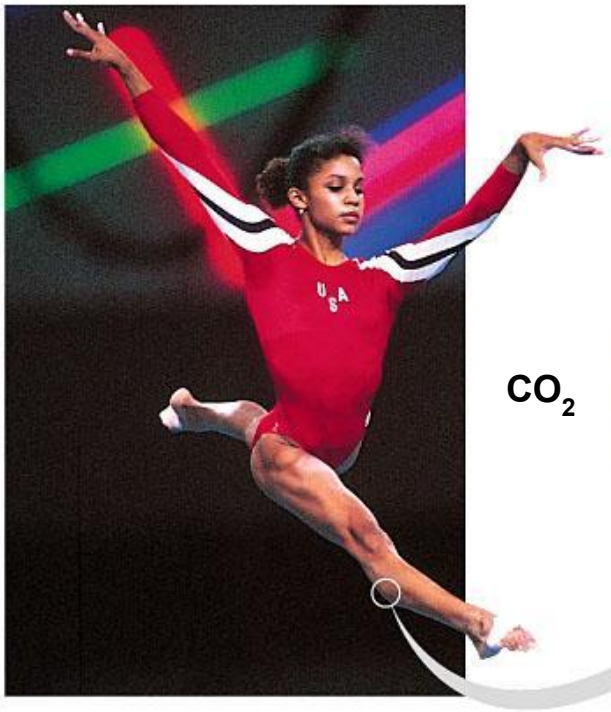
- Some cells have specific nutrient requirements and cannot make some compounds, e.g.,

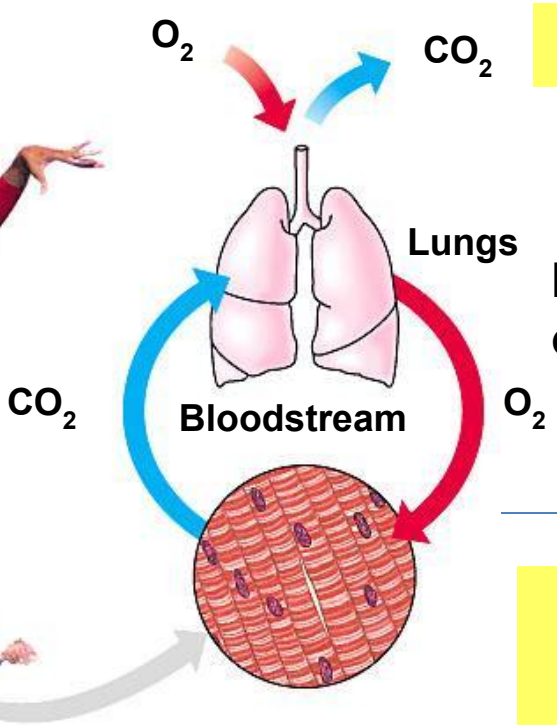
① vitamins
② some amino acids (about 1/2)
are required in the diet by man

– Some General Principles –

- Processes of metabolism are highly controlled
- Anabolism and catabolism are ***not*** **necessarily balanced** - one or the other may predominate in certain cells or at different times depending on cell needs
- The pathway to synthesize a complex substance is ***not*** **simply the reverse** of the degradative pathway.

Cellular Respiration





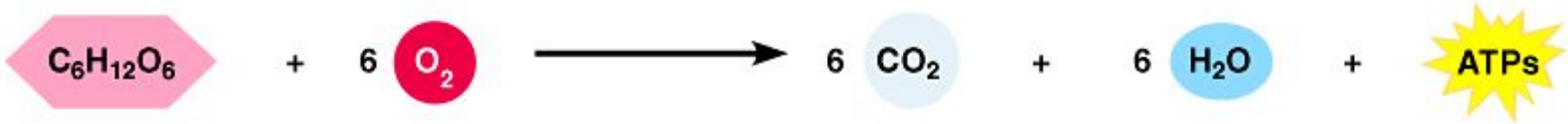
BREATHING

Muscle cells carrying out

CELLULAR RESPIRATION

$$\text{Sugar} + O_2 \rightarrow \text{ATP} + CO_2 + H_2O$$

Cellular Respiration uses oxygen and glucose to produce Carbon dioxide, water, and ATP.

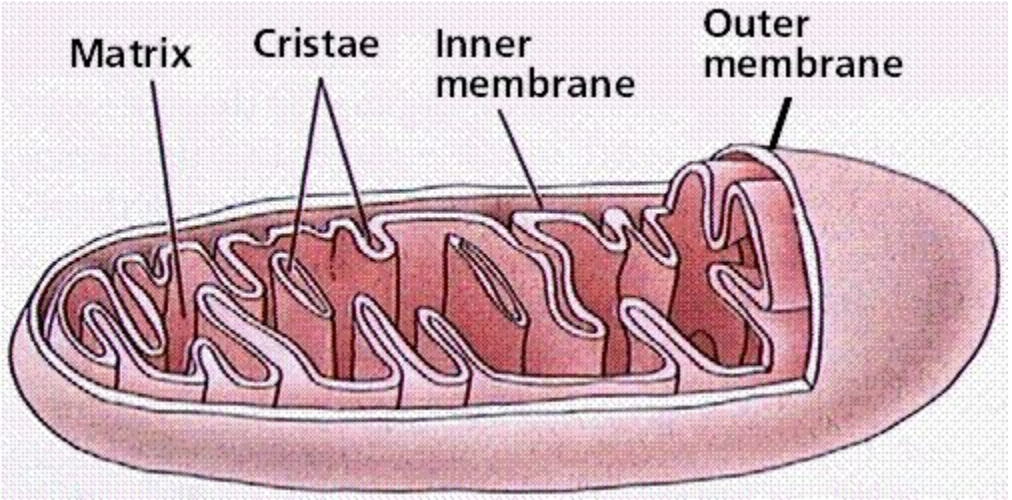


Glucose | Oxygen gas | Carbon dioxide | Water | Energy

Cellular Respiration – the transfer of energy from an organic compound into ATP

What are the Stages of Cellular Respiration?

- **STEP 1 - Glycolysis**
- **STEP 2 - The Krebs Cycle**
- **STEP 3 - The Electron Transport Chain (ETC)**



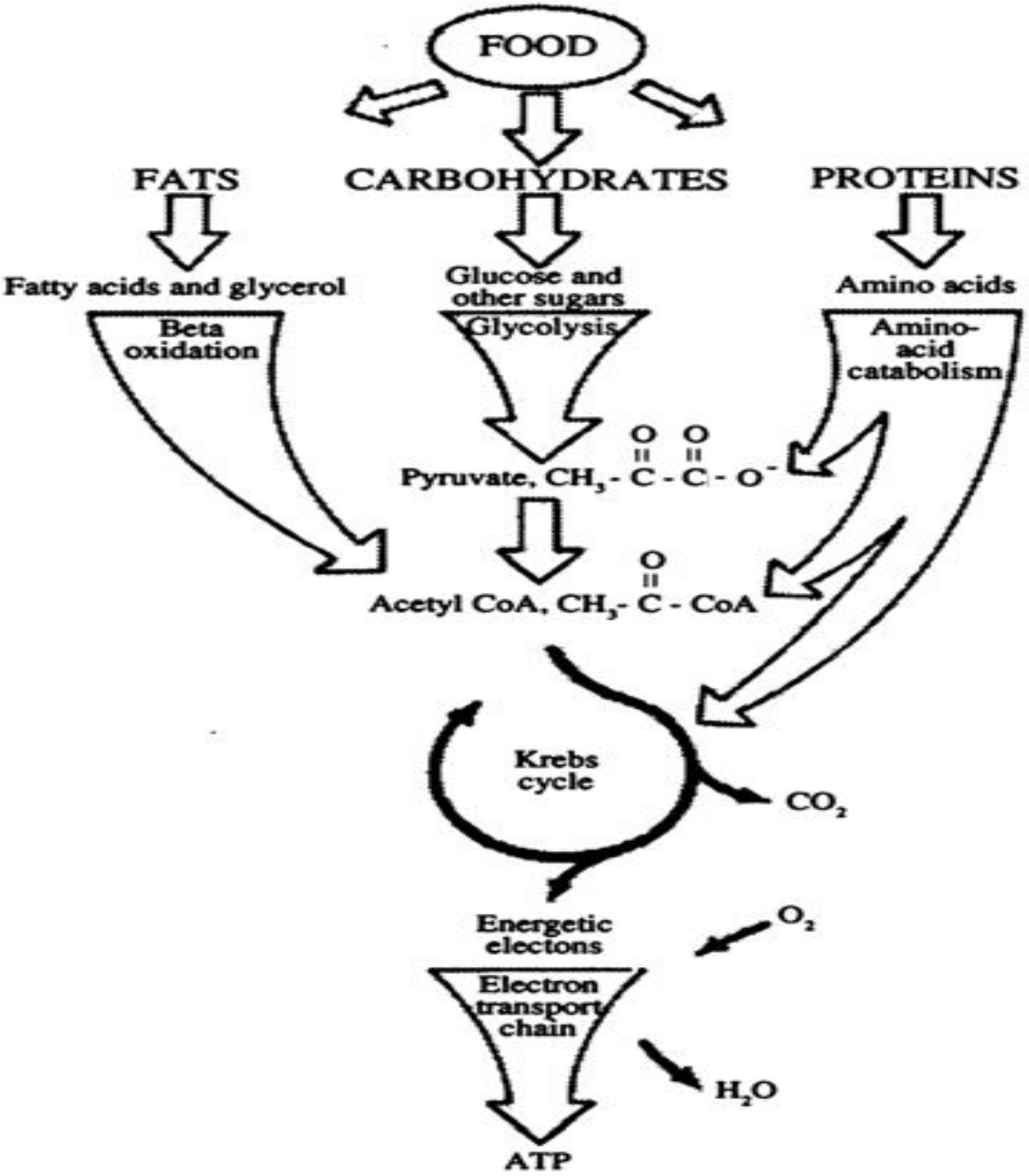
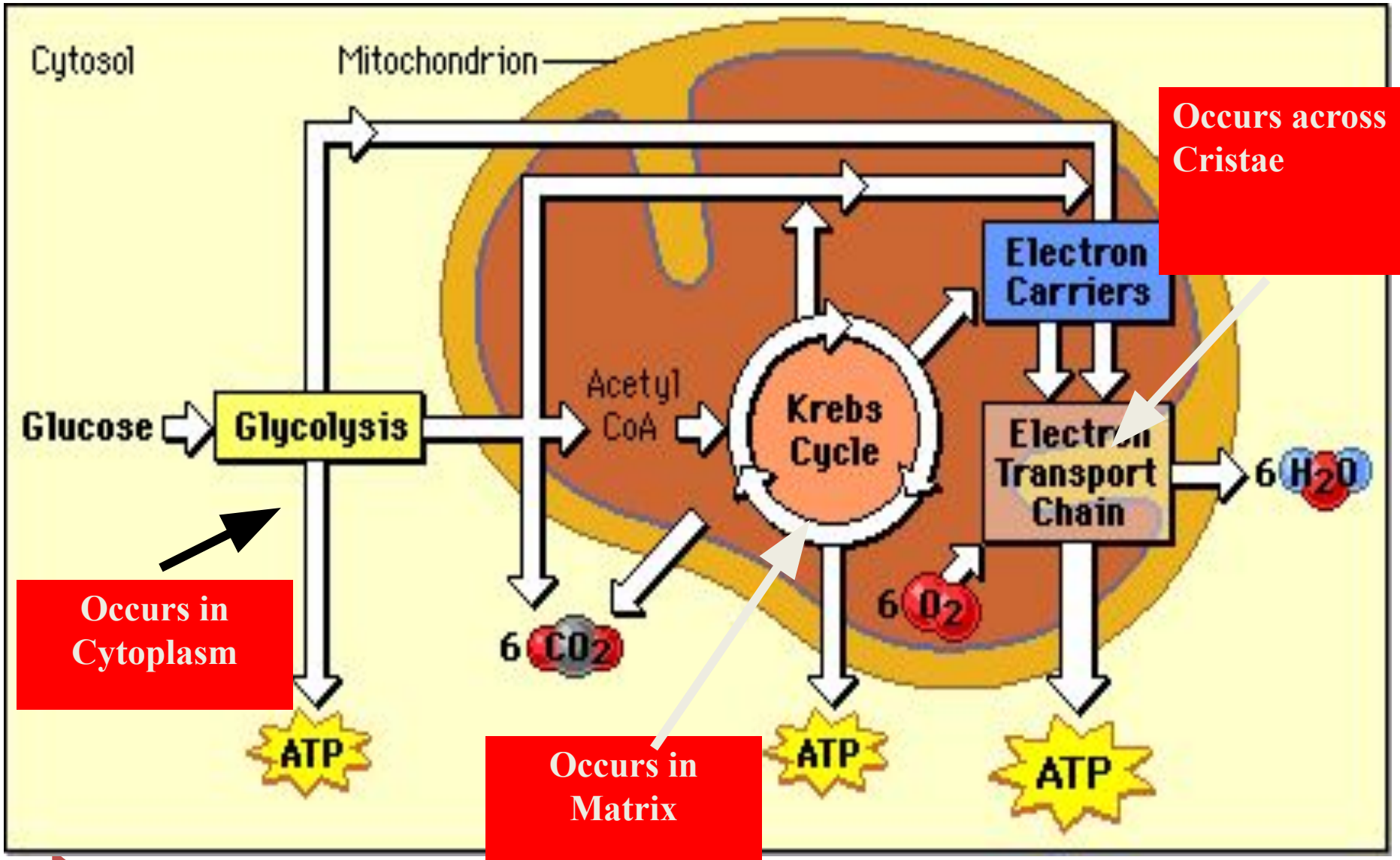
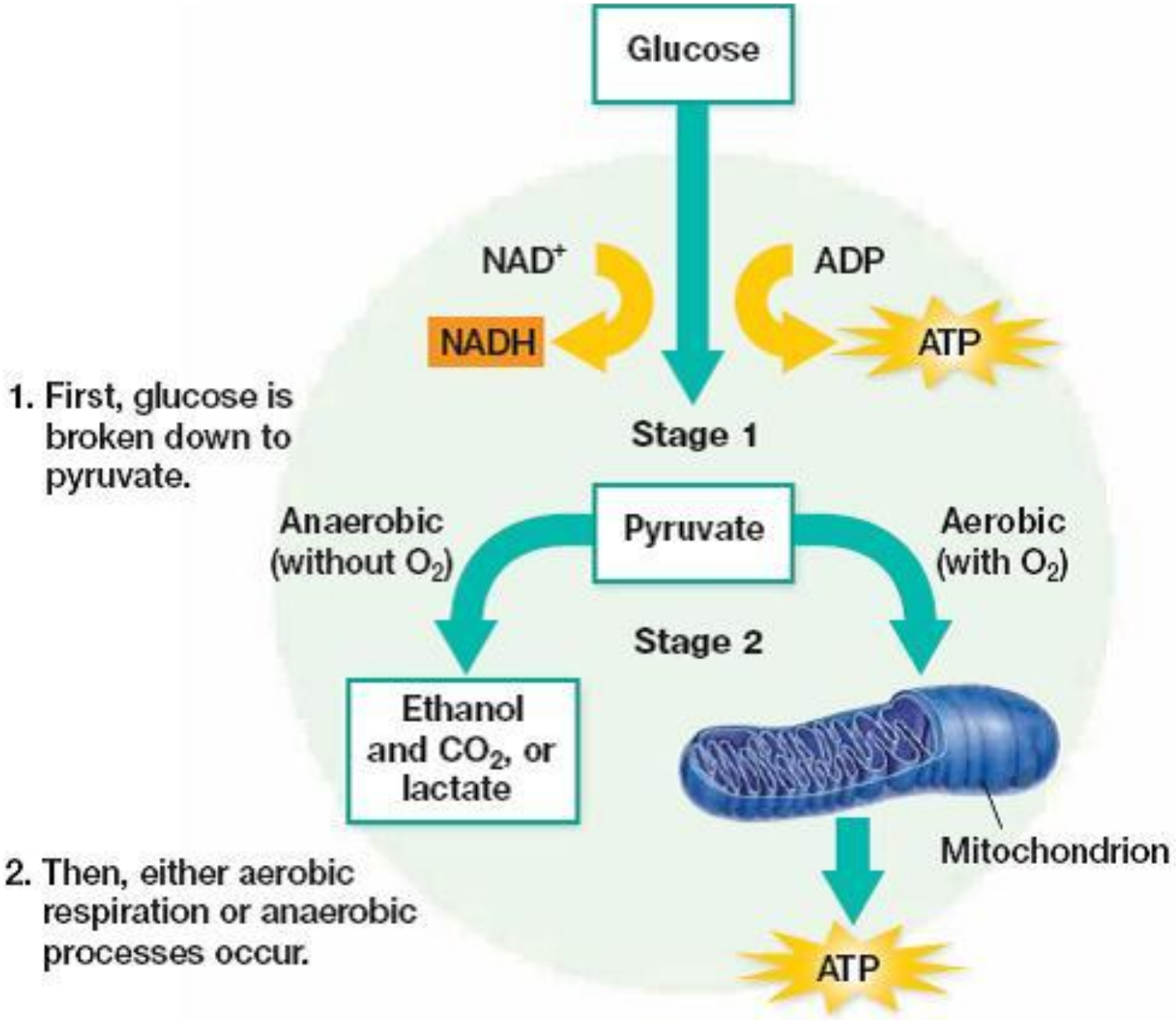

FOOD
FATS
CARBOHYDRATES
PROTEINS
Fatty acids and glycerol
Glucose and other sugars
Amino acids
Beta oxidation
Glycolysis
Amino-acid catabolism
Pyruvate, CH3 - C - C - O-
Acetyl CoA, CH3 - C - CoA
Krebs cycle
CO2
Energetic electons
O2
Electron transport chain
H2O
ATP

Diagram of the Process



Breakdown of Glucose



Important Molecules in Cellular Respiration

- NAD – Nicotinamide adenine dinucleotide

$NAD^+ + 2\,H \Rightarrow NADH + H^+$

- FAD – Flavin adenine dinucleotide

$FAD + 2\,H \Rightarrow FADH_2$

- CO_2 – Carbon Dioxide

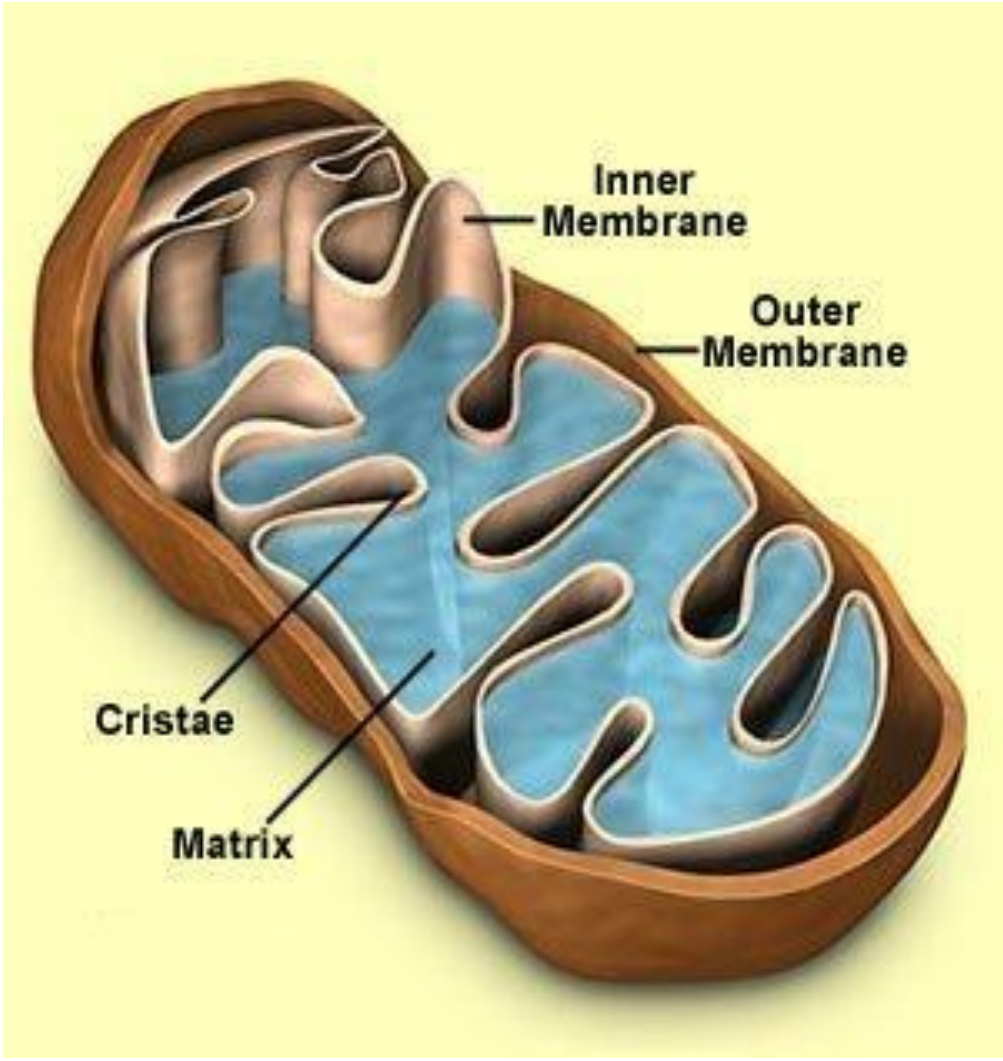
- ATP – Adenosine Triphosphate

Remember that 2H = 2 electrons and 2H+

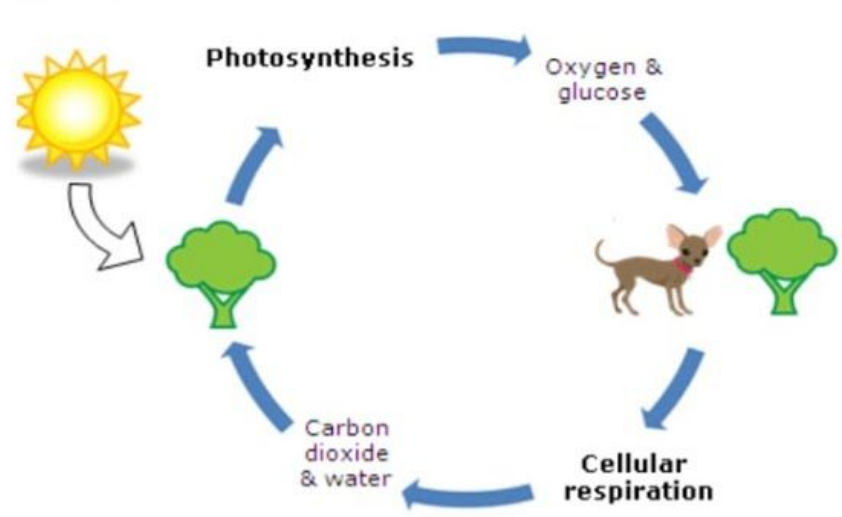
MITOCHONDRIA

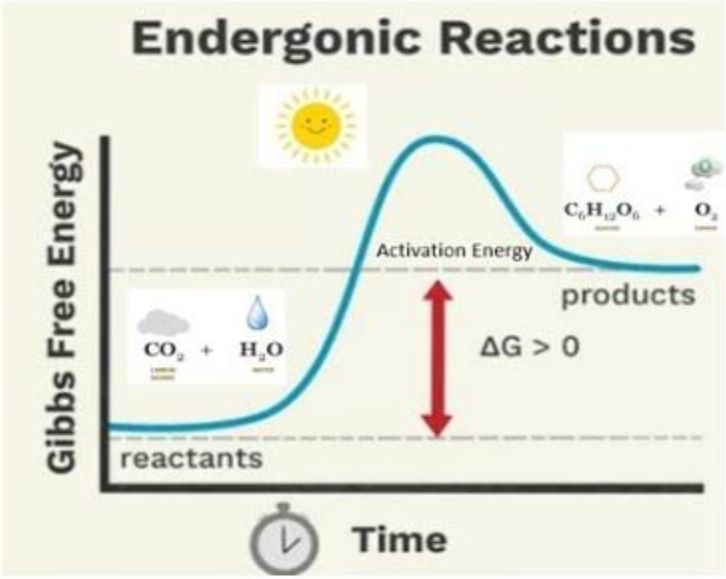
Krebs Cycle & ETC Take place in the Mitochondria →

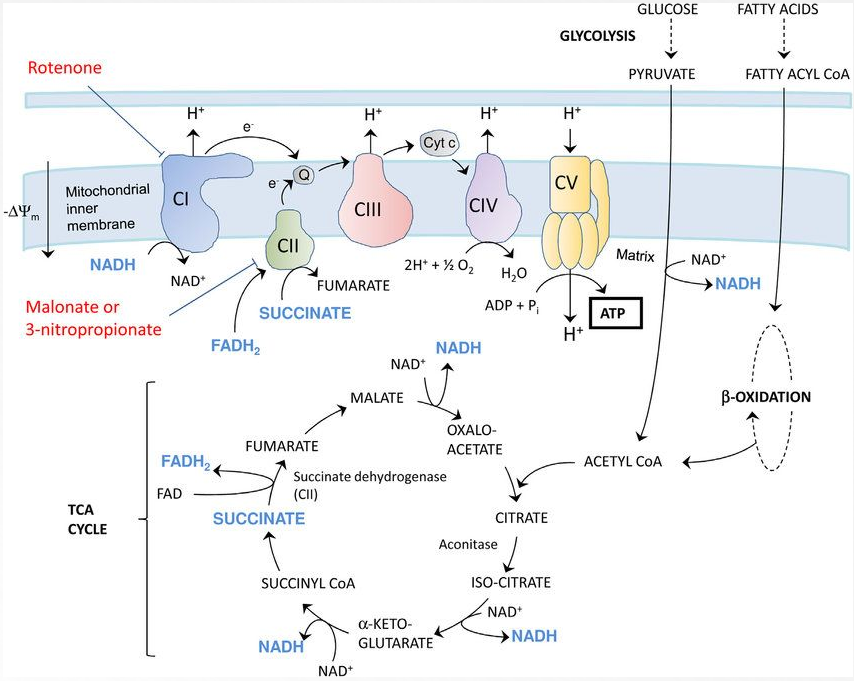
- **Smooth** outer Membrane
- **Folded** inner membrane
- Folds called **Cristae**
- Space inside cristae called the **Matrix, Gel like, 50% protein**

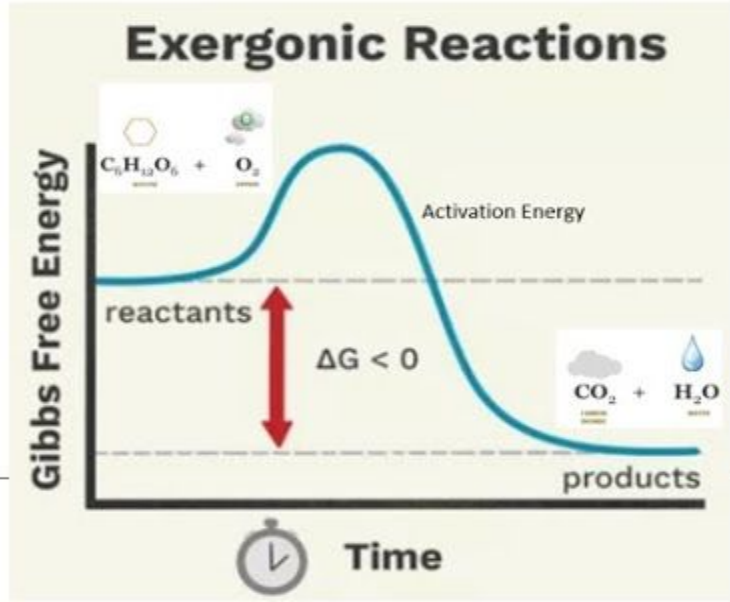


Bioenergetics & Electron Transport Chain





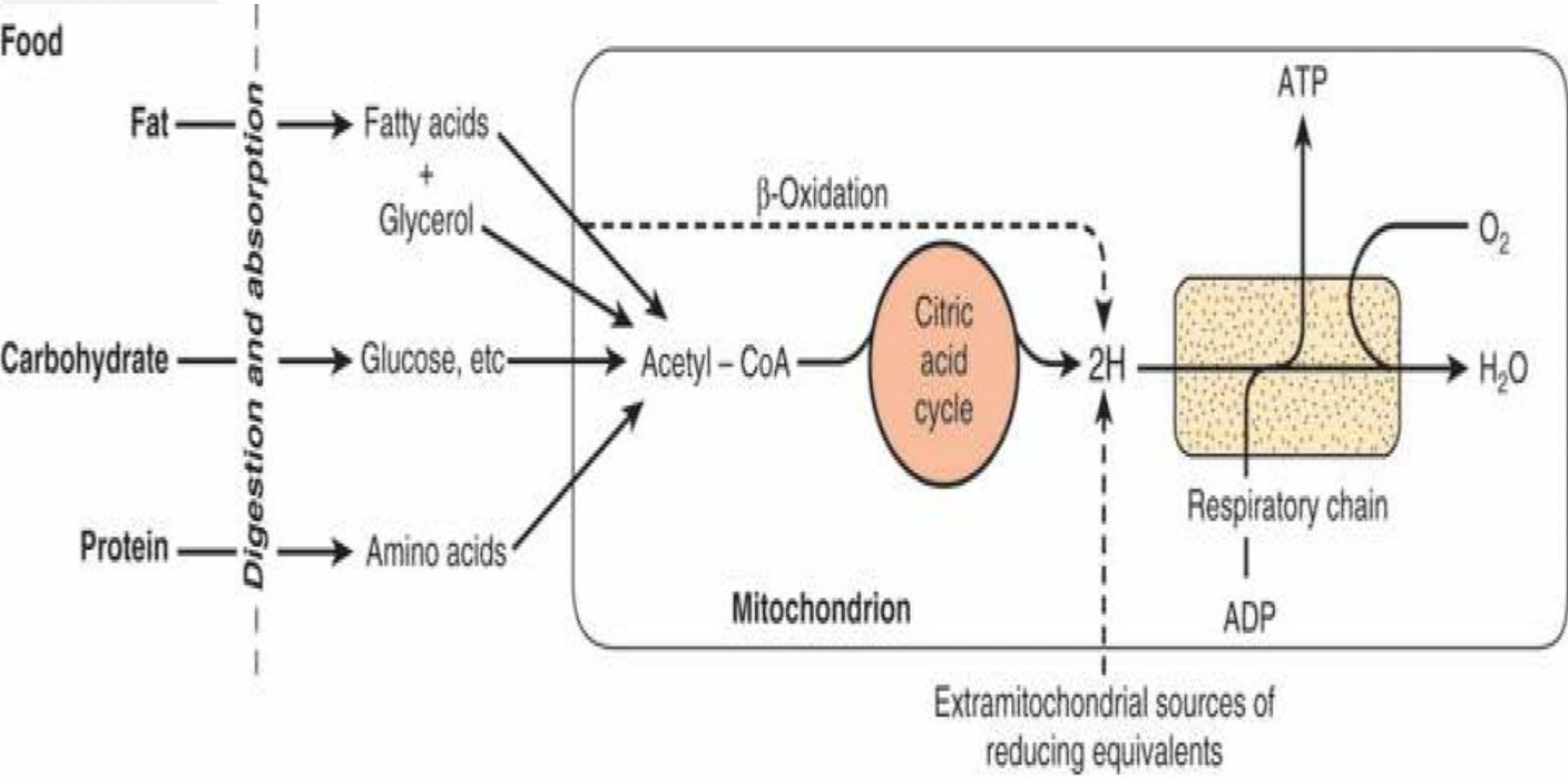




Learning Objectives

At the end of this lecture, students would be able to learn different concepts of:

- ✔ Fundamentals of Bioenergetics and Energy Production.
- ✔ Structure and Function of the Electron Transport Chain (ETC).
- ✔ ATP Synthesis and Oxidative Phosphorylation.


Food
Fat
Carbohydrate
Protein
Digestion and absorption
Fatty acids
+
Glycerol
Glucose, etc
Amino acids
β-Oxidation
Acetyl – CoA
Citric acid cycle
2H
ATP
O_2
H_2O
Respiratory chain
ADP
Mitochondrion
Extramitochondrial sources of reducing equivalents

Bioenergetics

- **Bioenergetics** or biochemical thermodynamics, is the study of the energy changes accompanying biochemical reactions.
- Three fundamental thermodynamic variables:
  - **Enthalpy (H):**
    - The heat content of physical object or body (system)
      - Derived from first law of thermodynamics.

- Change in enthalpy (ΔH) (Kcal/mol) is the heat absorbed or released during a reaction.

- Entropy (S):
- The randomness or disorder of a system.
- Derived from second law of thermodynamics.

- **Change in entropy (ΔS) is the degree of randomness or disorders created during the reaction.**
- **Free energy (G):**
- **The maximum usable work that can be obtained from a system at constant pressure, temperature and volume.**

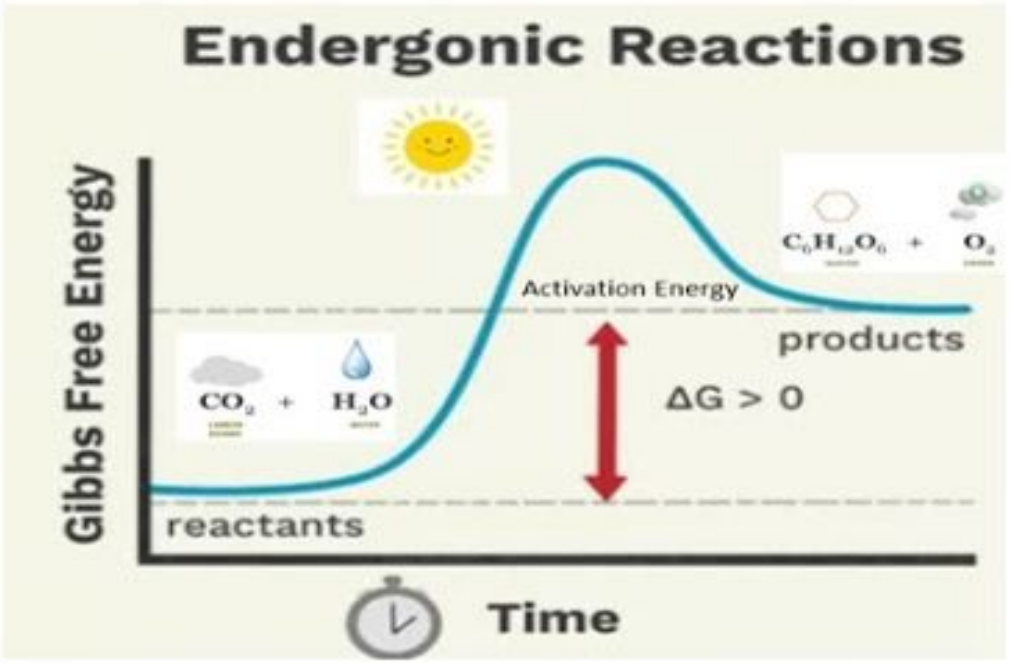
- **Free energy change (ΔG) is the change in free energy occurring during biological reactions.**
- **It is related to enthalpy & entropy**
- **Change in free energy can be expressed as:**
- **$\Delta G = \Delta H - T \Delta S$**
- **ΔH is the change in enthalpy**
- **ΔS is the change in entropy**
- **T is the absolute temperature**

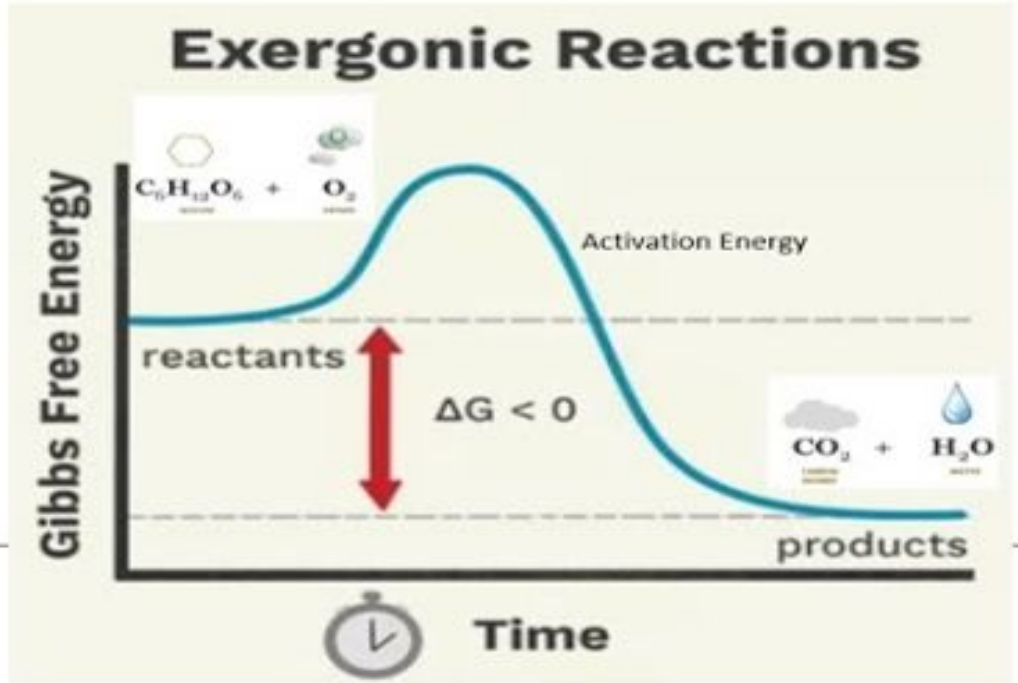
- **Standard free energy change (ΔG):**
- **It is defined as free energy change under standard conditions.**
- **Standard condition is defined as pH 7.0, temperature 25 $\mathring{C}$, all reactant concentration at 1m conc, all gases at pressure 1 atmosphere.**

Exergonic & Endergonic Reactions

- **Exergonic reactions:**
- **If the free energy change ΔG is negative in sign, the reaction proceeds spontaneously with loss of free energy & it is exergonic**
- **Exergonic is usually by breaking the bonds.**

- **Endergonic reactions:**
- **If the free energy change ΔG is positive, the reaction proceeds only if free energy can be gained & it is endergonic.**
- **Endergonic is usually by formation of the bonds.**
- **Reactions at equilibrium: If the free energy change is zero, the reaction is at equilibrium.**


Endergonic Reactions
Gibbs Free Energy
$C_6H_{12}O_6$ + O_2
Activation Energy
products
CO_2 + H_2O
ΔG > 0
reactants
Time


Exergonic Reactions
Gibbs Free Energy
$C_6H_{12}O_6$ + O_2
Activation Energy
reactants
ΔG < 0
CO_2 + H_2O
products
Time

Oxidation-Reduction Reactions

- Oxidation is defined as loss of electrons
- Loss of electrons occurs in three ways
  - (1) Direct loss of electrons
  - (2) Removal of hydrogen
  - (3) Addition of oxygen
- Electrons are transferred as
  - (1) Hydride ions(H^-)
  - (2) Hydrogen atoms (H)
  - (3) Electrons (e^-)

- **Reduction:**
- **Reduction is defined as the gain of electrons.**
- **Eg: ferric iron(Fe^{3+}) to ferrous iron (Fe^{2+})**
- **Oxidation-Reduction Reactions:**
- **Oxidation-reduction reactions involve transfer of electrons from one compound to another.**
- **When one substrate is oxidized, another substrate is simultaneously reduced.**

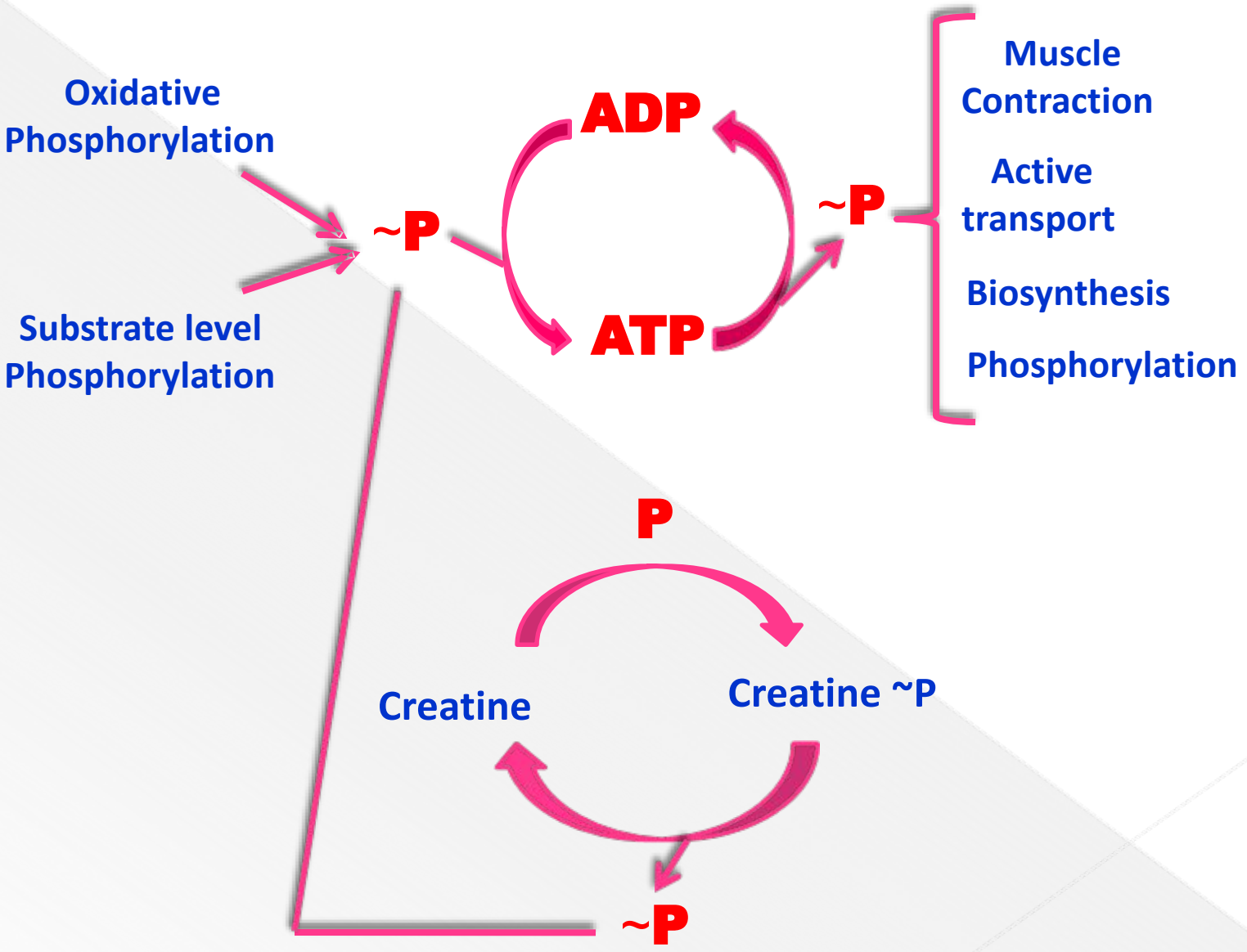
ATP-ADP Cycle

- The hydrolysis of ATP is associated with the release of large amount of energy.

$$ATP + HO_2 \longrightarrow ADP + Pi + 7.3\ Cal$$

- The energy liberated is utilized for various process like muscle contraction, active transport etc.
- ATP can also acts as a donor of high-energy phosphate to low-energy compounds, to make them energy rich.
- ADP can accepts phosphate to form ATP.

ATP-ADP cycle along with sources & utilization of ATP



- ATP serves as an immediately available energy currency of the cell which is constantly being utilized & regenerated.
- ATP acts as an energy link between the catabolism & anabolism in the biological systems.
- Hydrolysis of ATP releases 7.3kcal/mol.

Synthesis of ATP

- ATP can be synthesized in two ways
- **Oxidative phosphorylation:**
- Major source of ATP in aerobic organisms.
- It is linked with mitochondrial ETC.
- **Substrate level phosphorylation:**
- When the energy of high energy compound is directly transferred to nucleoside diphosphate to form a triphosphate without the help from ETC.

- The high-energy compounds such as
- PEP
- 1,3-bisphosphoglycerate
- Succinyl CoA can transfer high-energy phosphate to ultimately produce ATP.

- **Storage forms:**
- **Phosphocreatine (creatine phosphate) provides high energy reservoir of ATP to regenerate ATP rapidly, catalyzed by creatine kinase.**
- **Stored mainly in muscle & brain.**
- **In invertebrates, phosphoarginine (arginine phosphate) is storage form.**

Biological oxidation

- The transfer of electrons from the reduced coenzymes through the respiratory chain to oxygen is known as biological oxidation.
- Energy released during this process is trapped as ATP.
- This coupling of oxidation with phosphorylation is called oxidative phosphorylation.

Transport of Reducing Equivalents- Shuttle Pathways

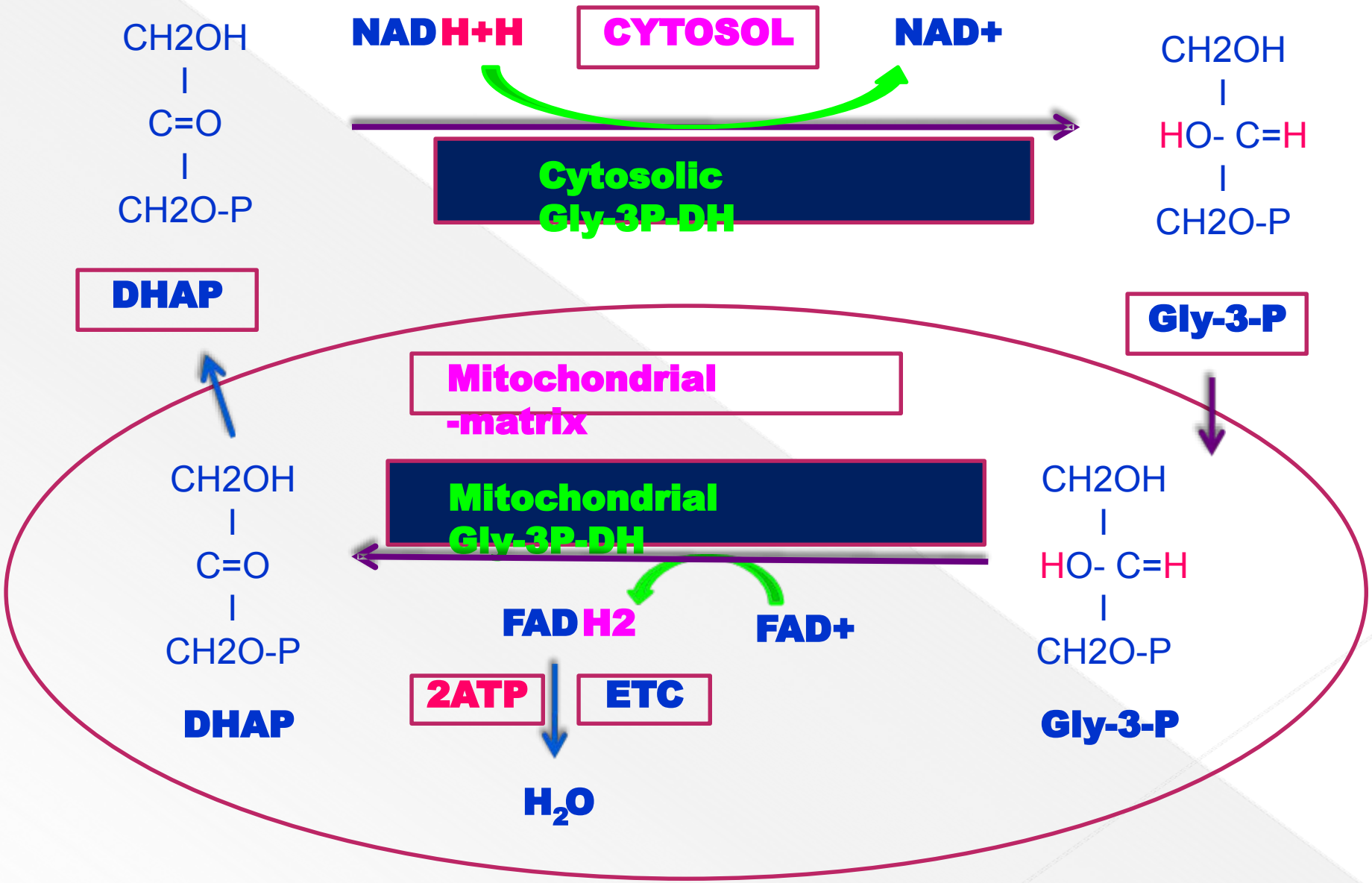
- The inner mitochondrial is impermeable to NADH.
- Therefore, the NADH produced in the cytosol cannot directly enter the mitochondria.
- Two pathways
- Glycerol-phosphate shuttle
- Malate-aspartate shuttle

Glycerol-phosphate shuttle

- Cytosolic glycerol 3-phosphate dehydrogenase oxidizes NADH to NAD^+
- The reducing equivalents are transported through glycerol 3-phosphate into the mitochondria.
- Glycerol 3-phosphate dehydrogenase-present on outer surface of inner mitochondrial membrane – reduces FAD to $FADH_2$.

- Dihydroxyacetone phosphate (DHAP) escapes into the cytosol & the shuttling continues.
- $FADH_2$ gets oxidized via ETC to generate 2ATP

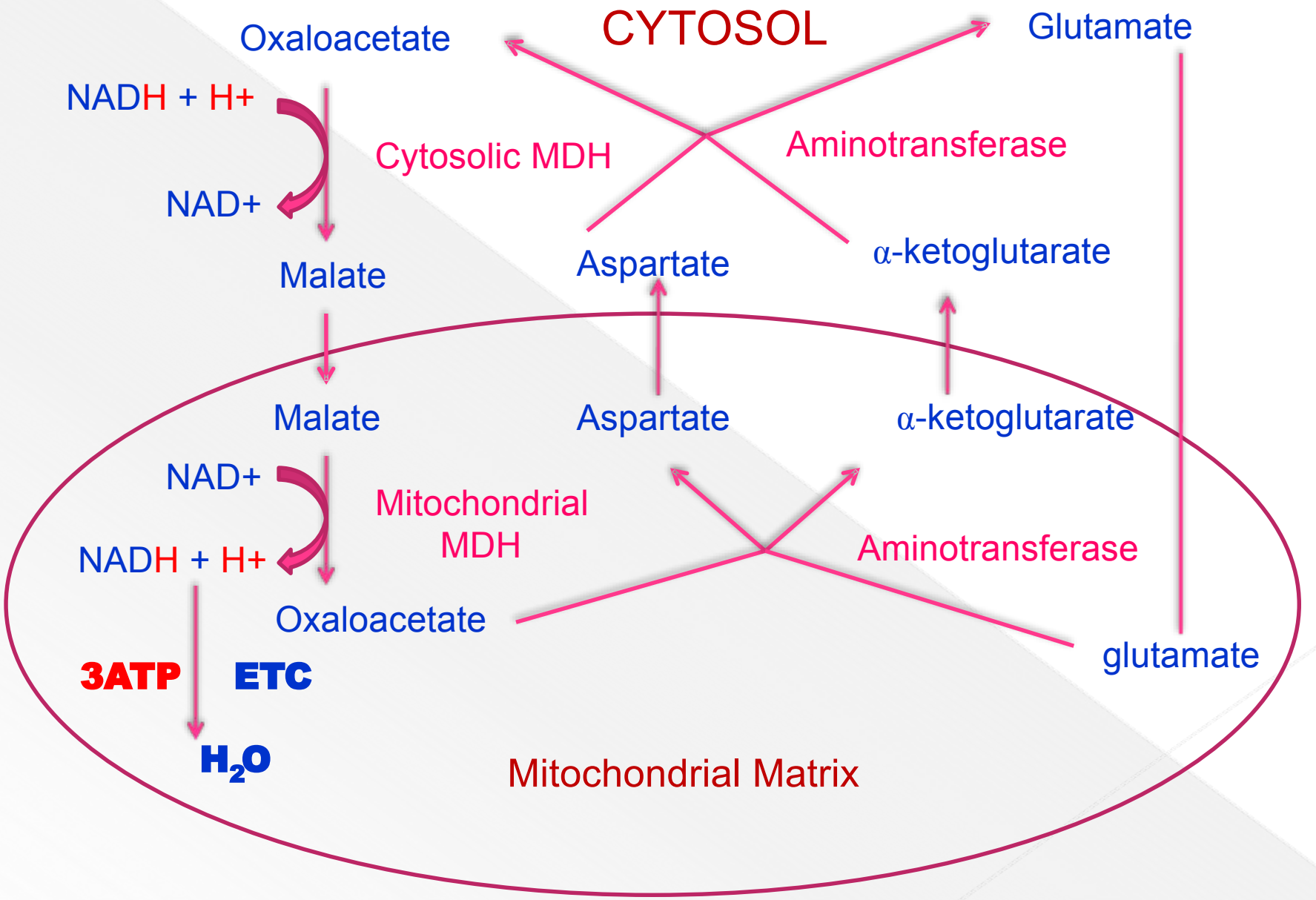
Glycerol-phosphate -shuttle



Malate-aspartate shuttle

- In the cytosol, oxaloacetate accepts the reducing equivalents (NADH) & becomes malate.
- Malate enters the mitochondria where it is oxidized by mitochondrial MDH
- In this reaction, NADH & oxaloacetate are regenerated.
- NADH gets oxidized via ETC & 3 ATP are produced.

Malate-aspartate shuttle



- In the mitochondria, oxaloacetate participates in transamination reaction with glutamate to produce aspartate & α-ketoglutarate.
- The aspartate enters the cytosol & transaminates with α-ketoglutarate to give oxaloacetate & glutamate.

Electron transport chain or respiratory chain

- The flow of electrons occurs through successive dehydrogenase enzymes in mitochondria , together known as the ETC.
  (the electrons are transferred from higher to lower potential.)
- **Significance:**
- The free energy released during the transport of electrons is utilized for the formation of ATP.

Mitochondrial organization

- Mitochondria consists of five distinct parts
- Outer membrane, inner membrane, intermembrane space, cristae & matrix
- Inner mitochondrial membrane:
- The ETC & ATP synthesizing system are located on inner mitochondrial membrane, which is specialized structure, rich in proteins.

- Inner membrane is highly folded to form cristae.
- Surface area of inner mitochondrial membrane is increased due to cristae.
- The inner surface of inner mitochondrial membrane possesses specialized particles, the phosphorylating subunits which are centres for ATP production.

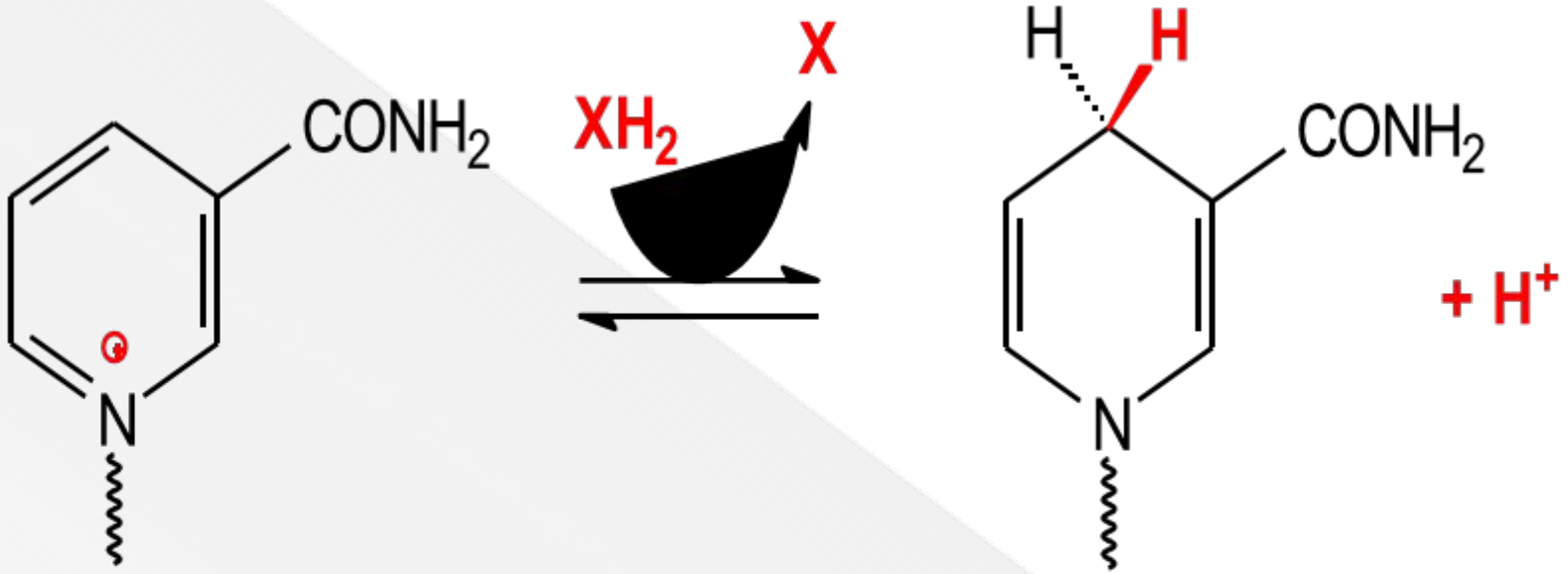
Components

- ETC consists of four enzymes complexes & two free electron carriers.
- Enzyme complexes:
- Complex I: NADH-ubiquinone oxido-reductase
- Complex II: Succinate dehydrogenase
- Complex III: Ubiquinol cytochrome oxido-reductase

- **Complex IV:** Cytochrome oxidase
- Two free electron carriers are coenzyme Q & Cytochrome C.
- **Complex V:** It is ATP synthase.
- The complexes I-IV are carriers of electrons while complex V is responsible for ATP synthesis.

- The enzyme complexes & mobile carriers are collectively involved in the transport of electrons which, ultimately, combine with oxygen to produce water.
- Largest proportion of O_2 supplied to body is utilized by mitochondria for the operation of ETC.

Nicotinamide coenzymes: NAD^+

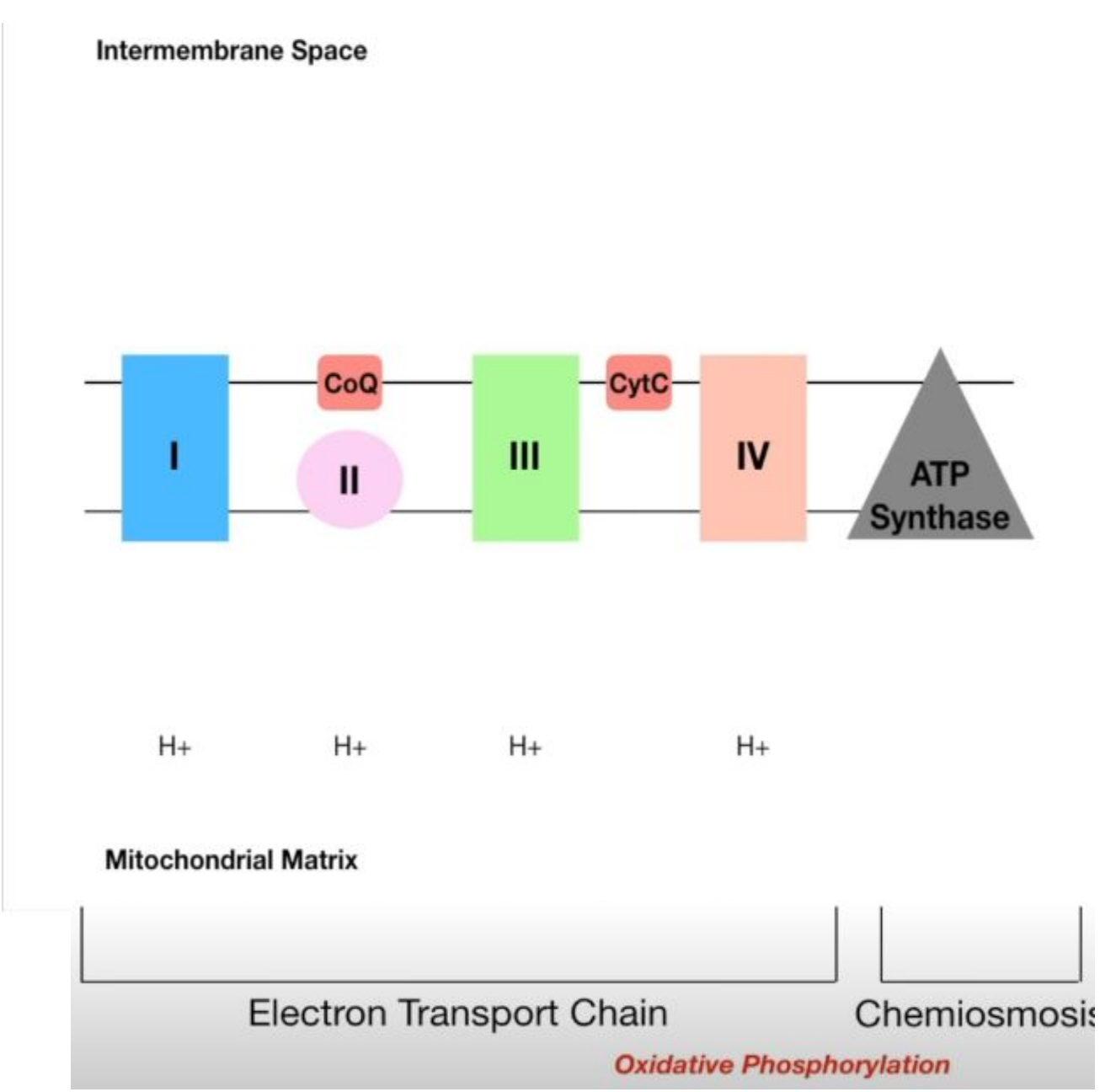


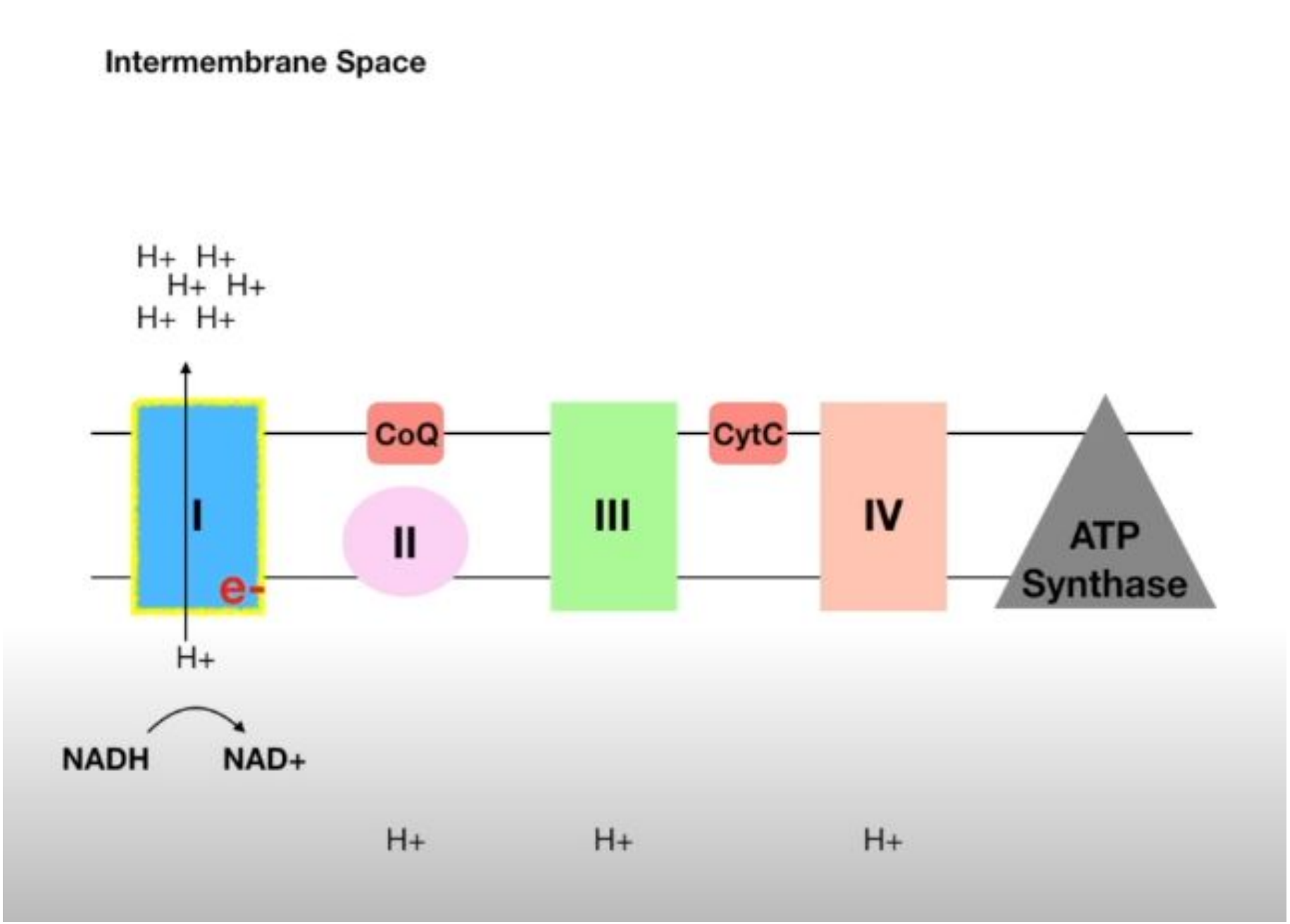
oxidised coenzyme

NAD^+ or $NADP^+$

reduced coenzyme

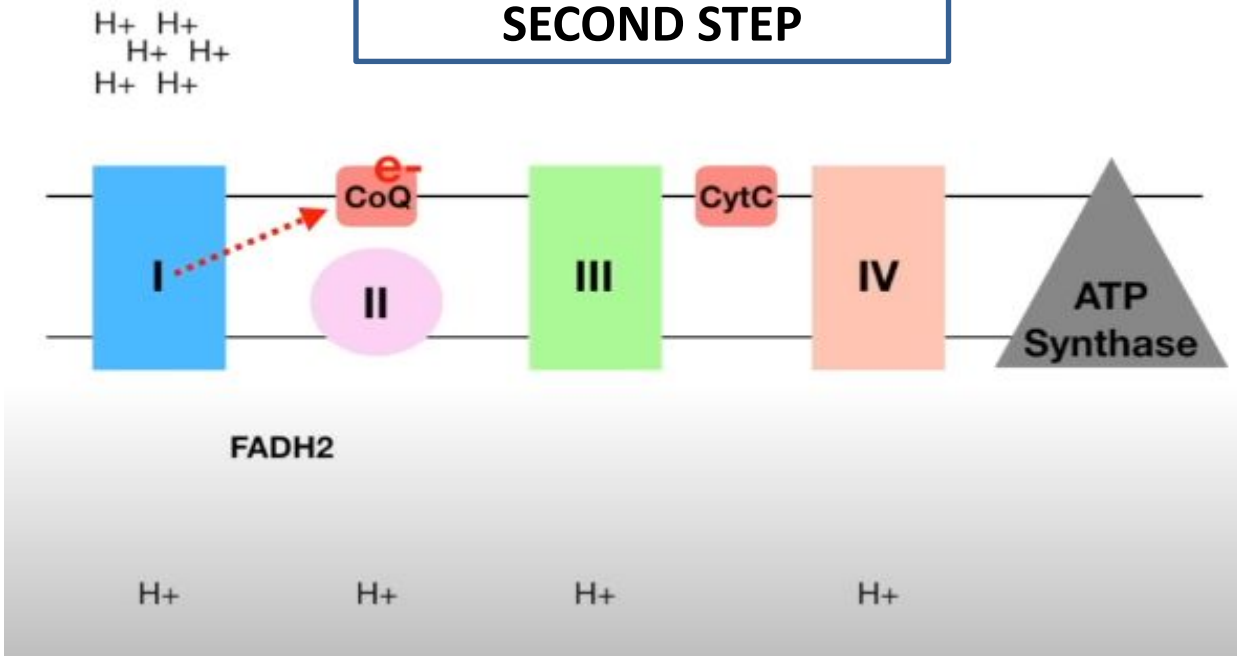
NADH or NADPH

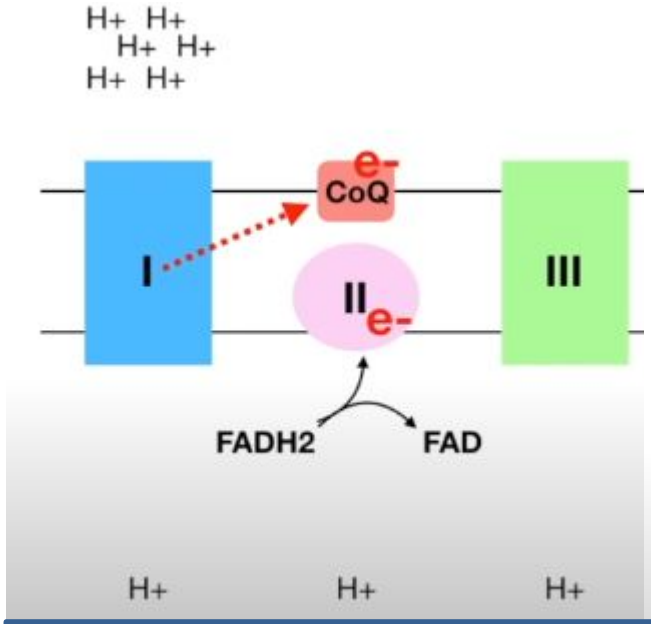

Intermembrane Space
I
CoQ
II
III
CytC
IV
ATP Synthase
H+
H+
H+
H+
Mitochondrial Matrix
Electron Transport Chain
Chemiosmosis
Oxidative Phosphorylation

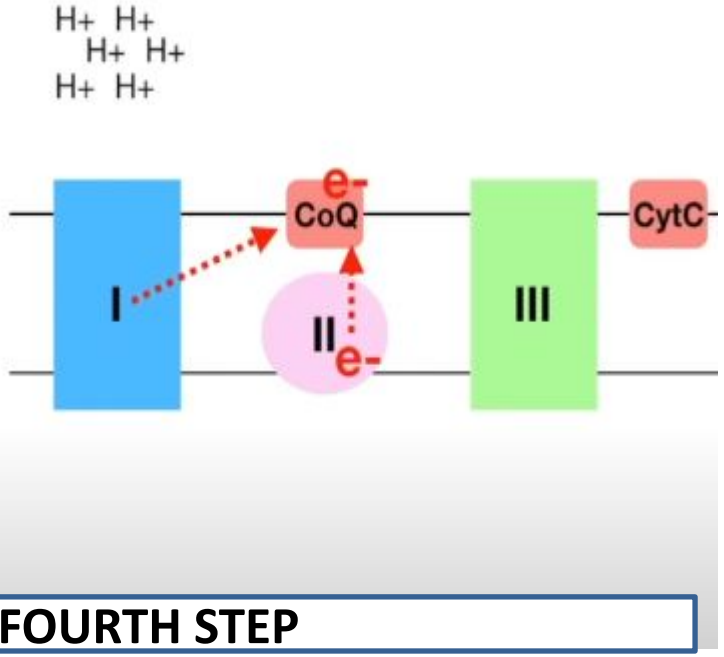

Intermembrane Space
H+ H+
H+ H+
H+ H+
I
e-
H+
CoQ
II
III
CytC
IV
ATP
Synthase
NADH
NAD+
H+
H+
H+

FIRST STEP

SECOND STEP

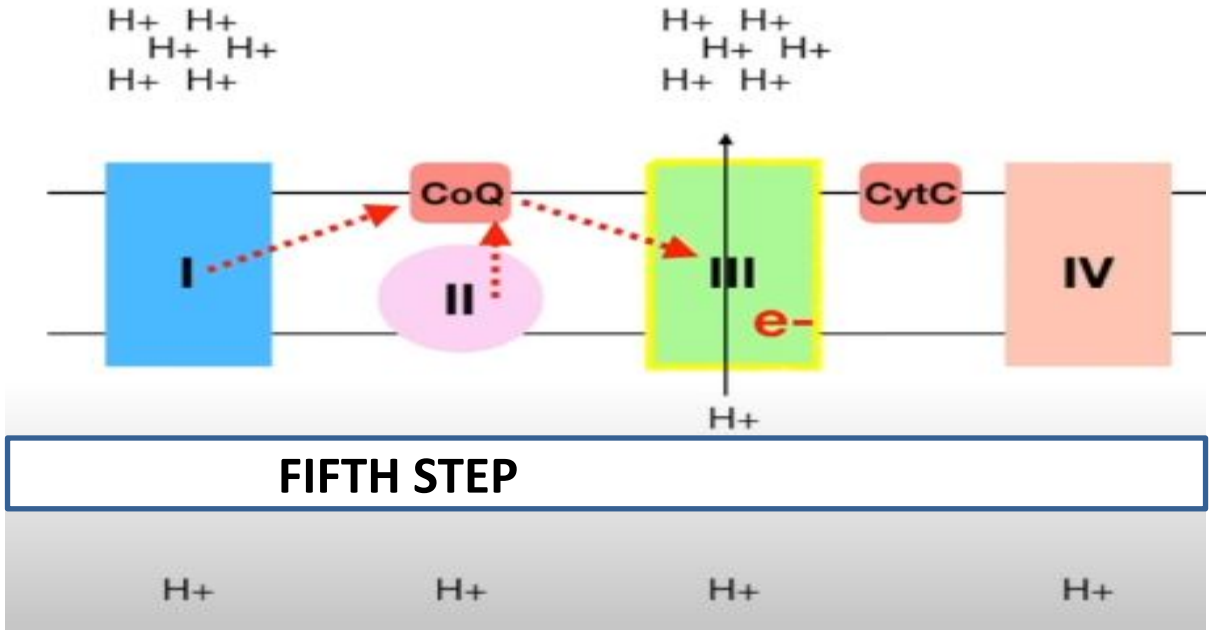




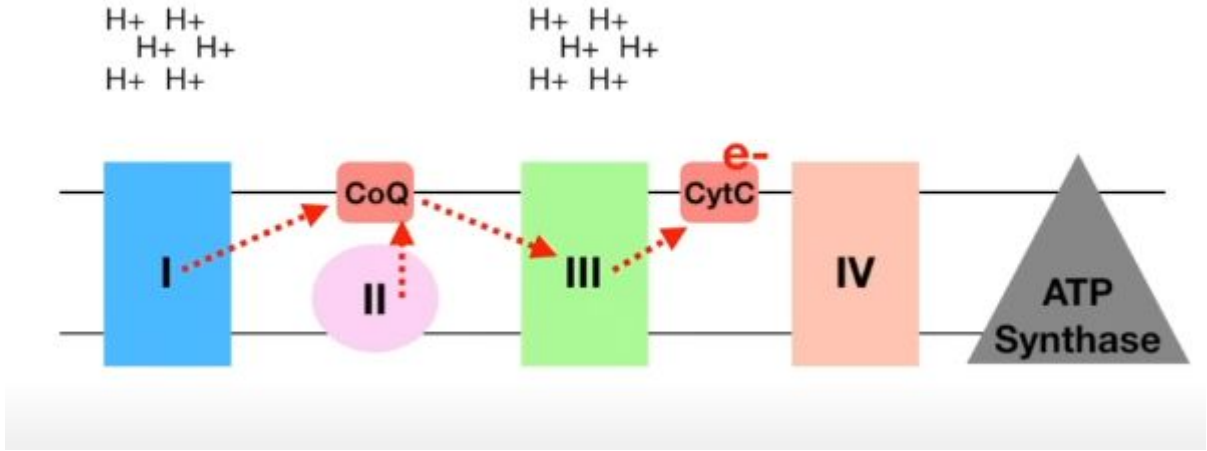


THIRD STEP

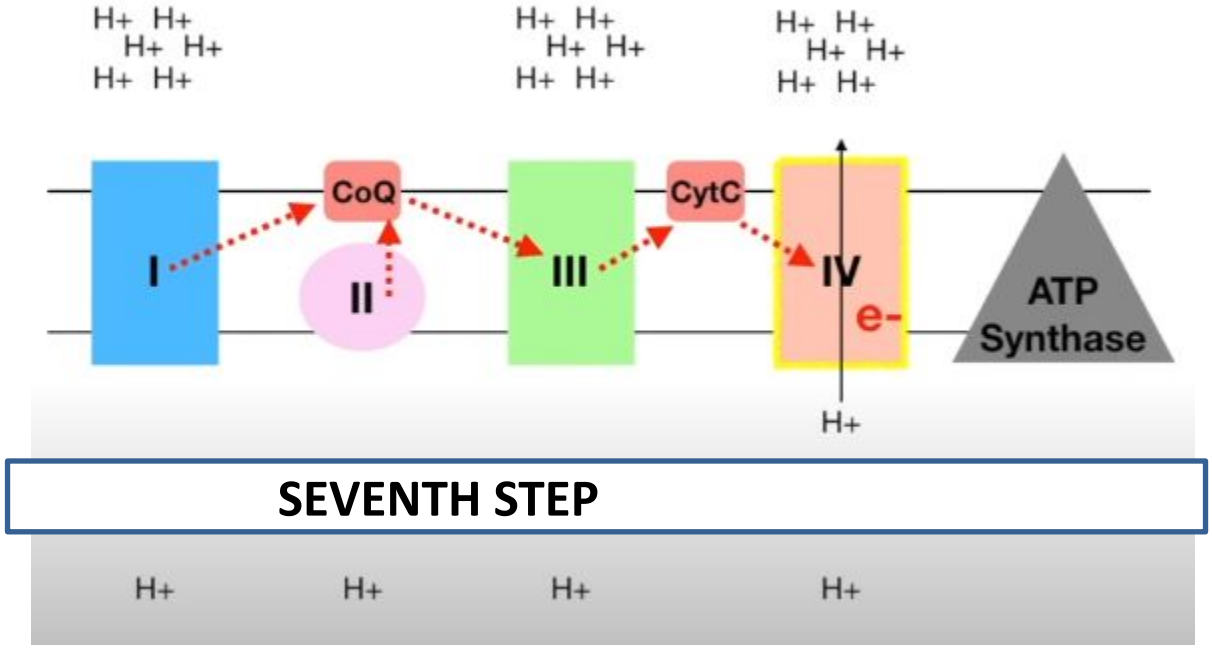
FOURTH STEP



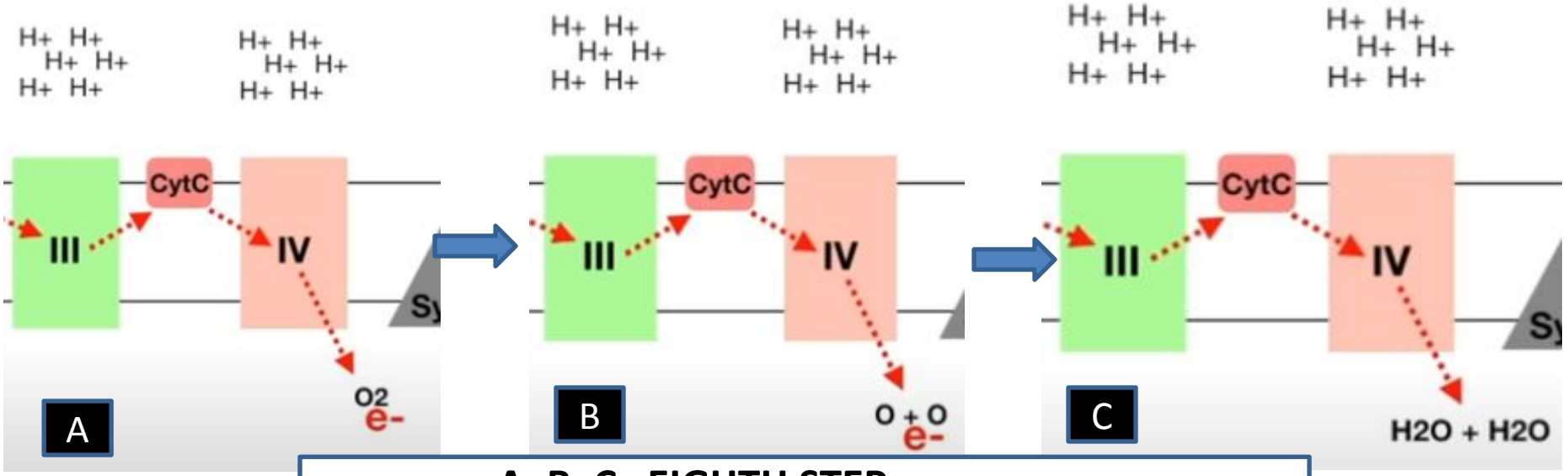
FIFTH STEP



SIXTH STEP



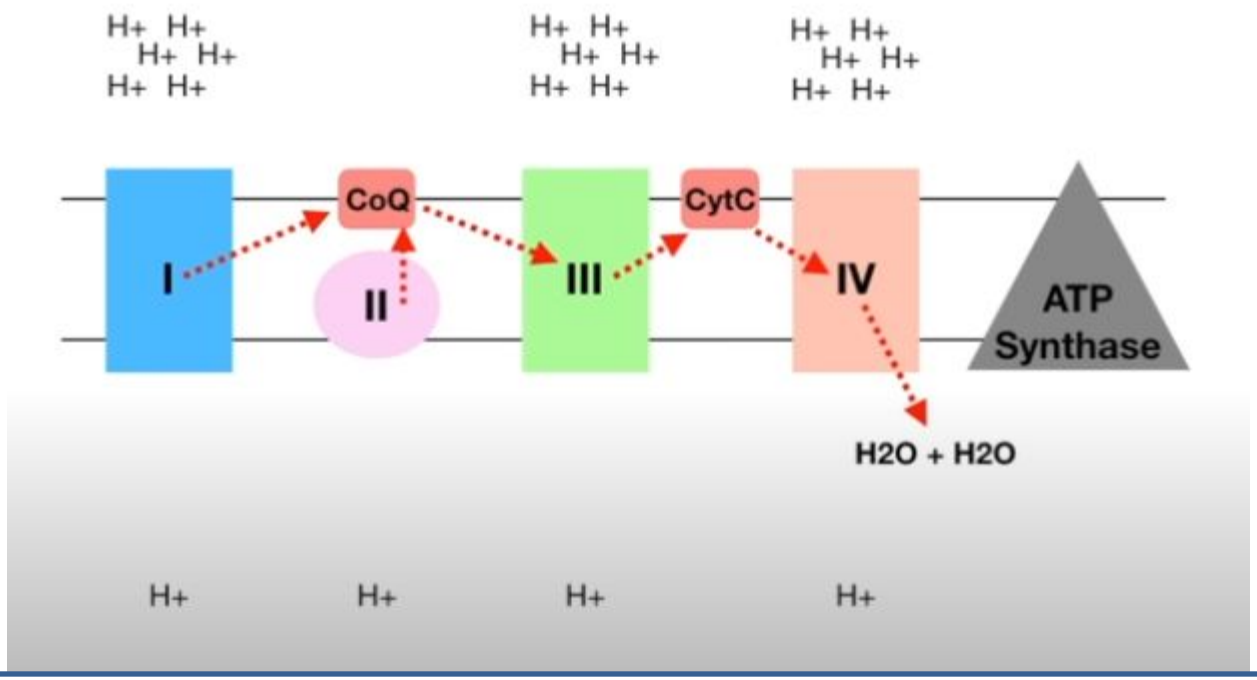
SEVENTH STEP



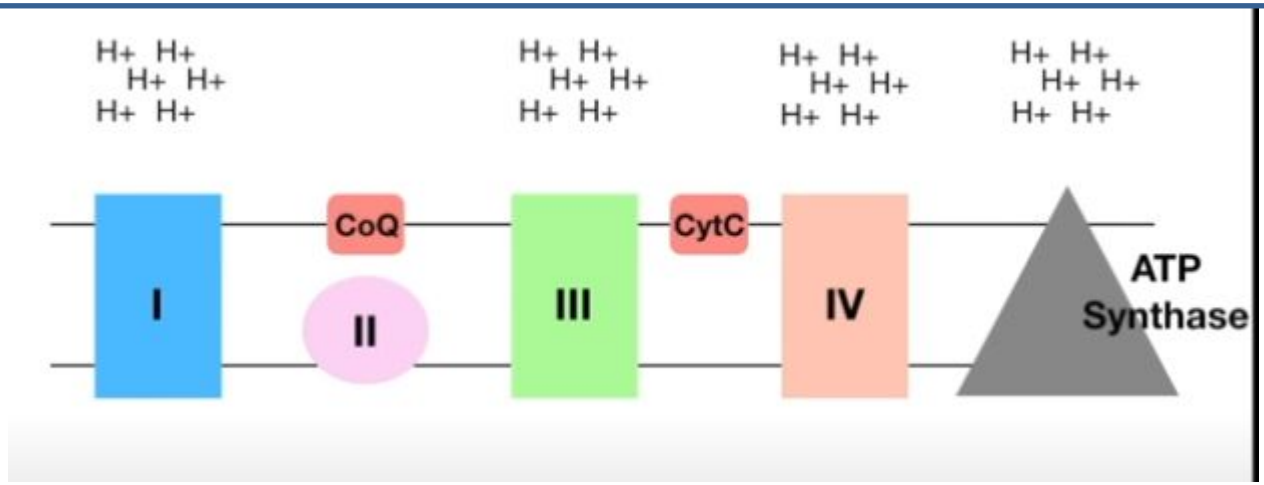
A, B, C EIGHTH STEP

Intermembrane Space

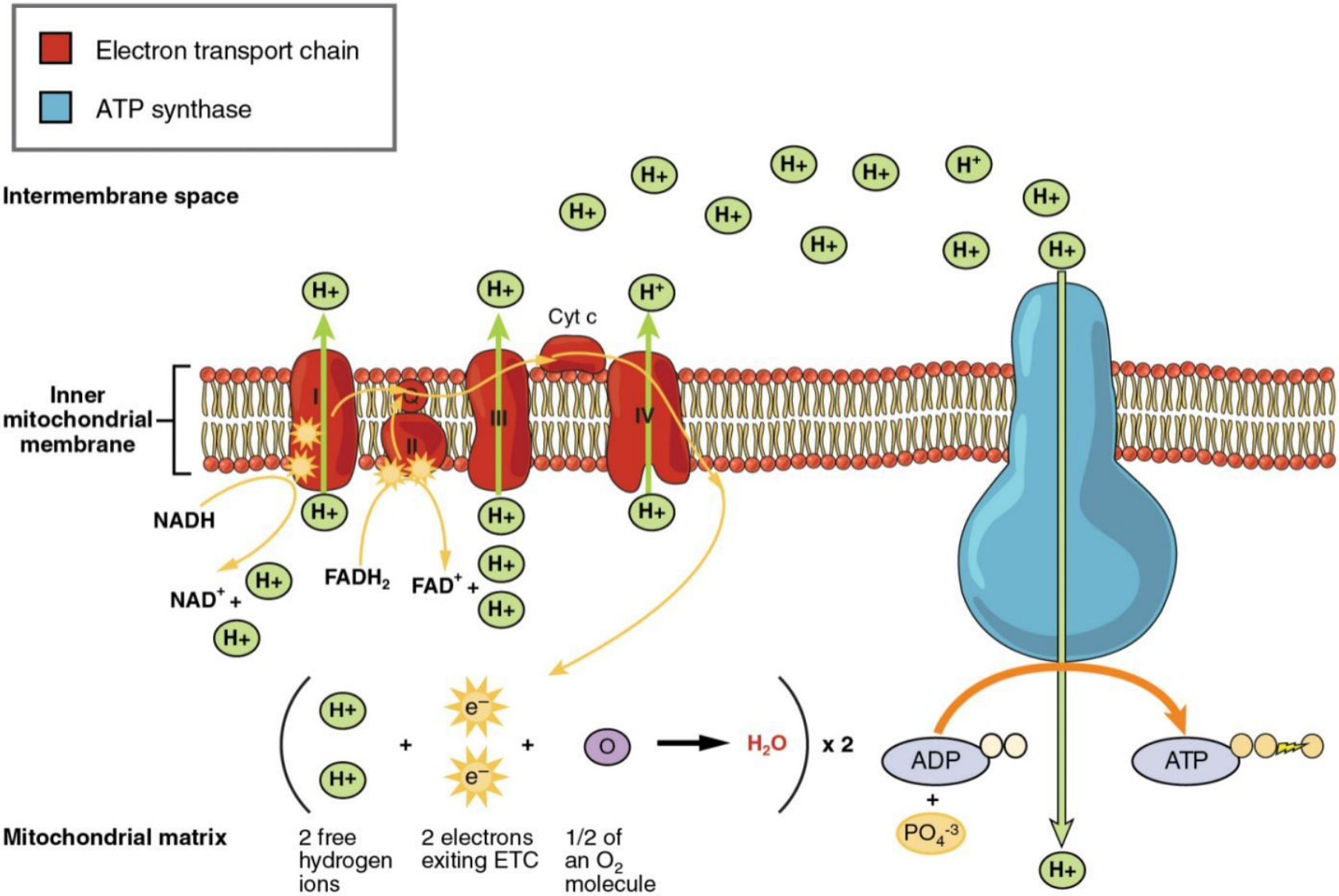
SUMMARY

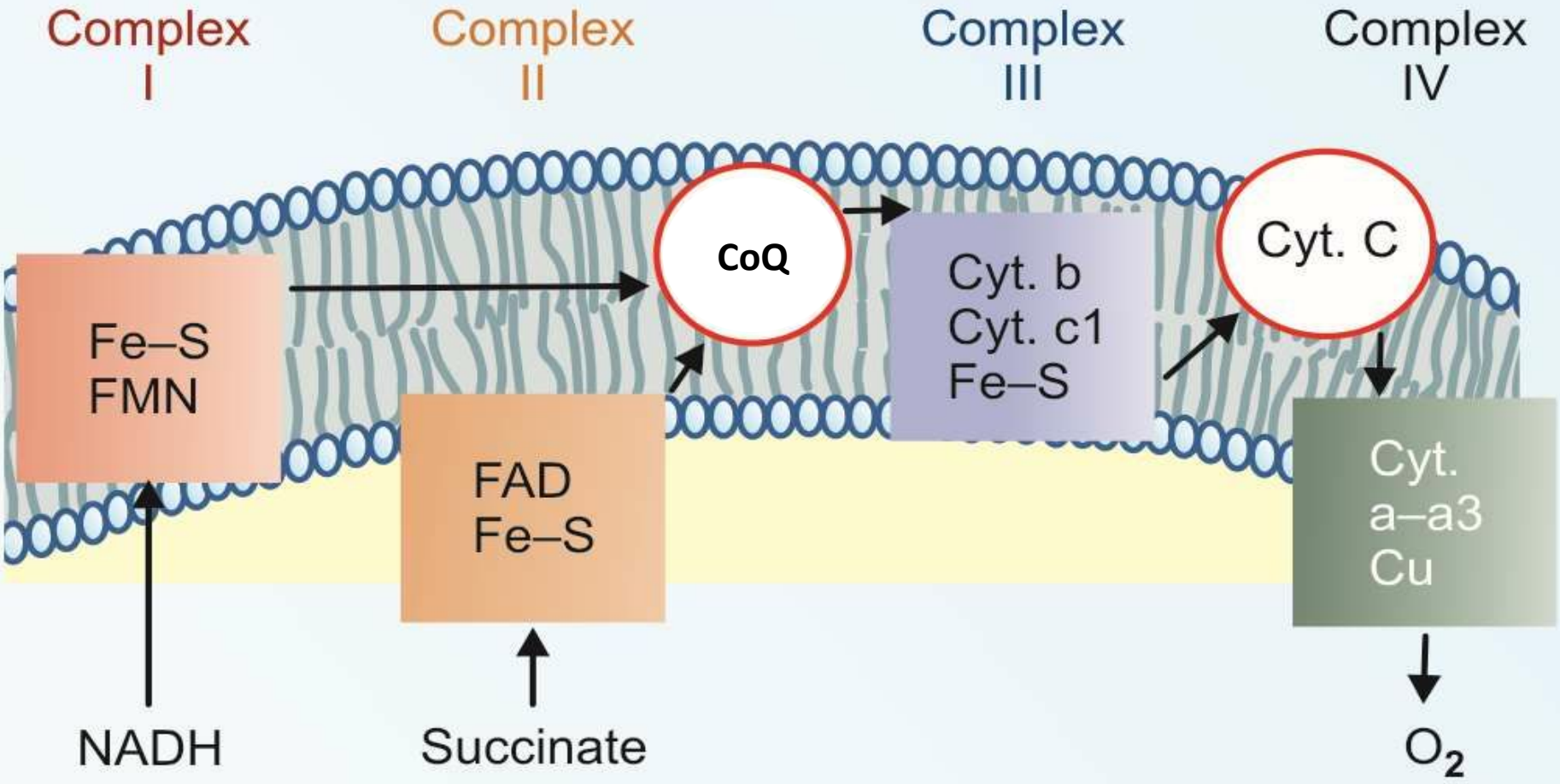


MORE PROTONS IN IMS THAN MATRIX CREATES ELECTROCHEMICAL GRADIENT



ATP SYNTHESIS



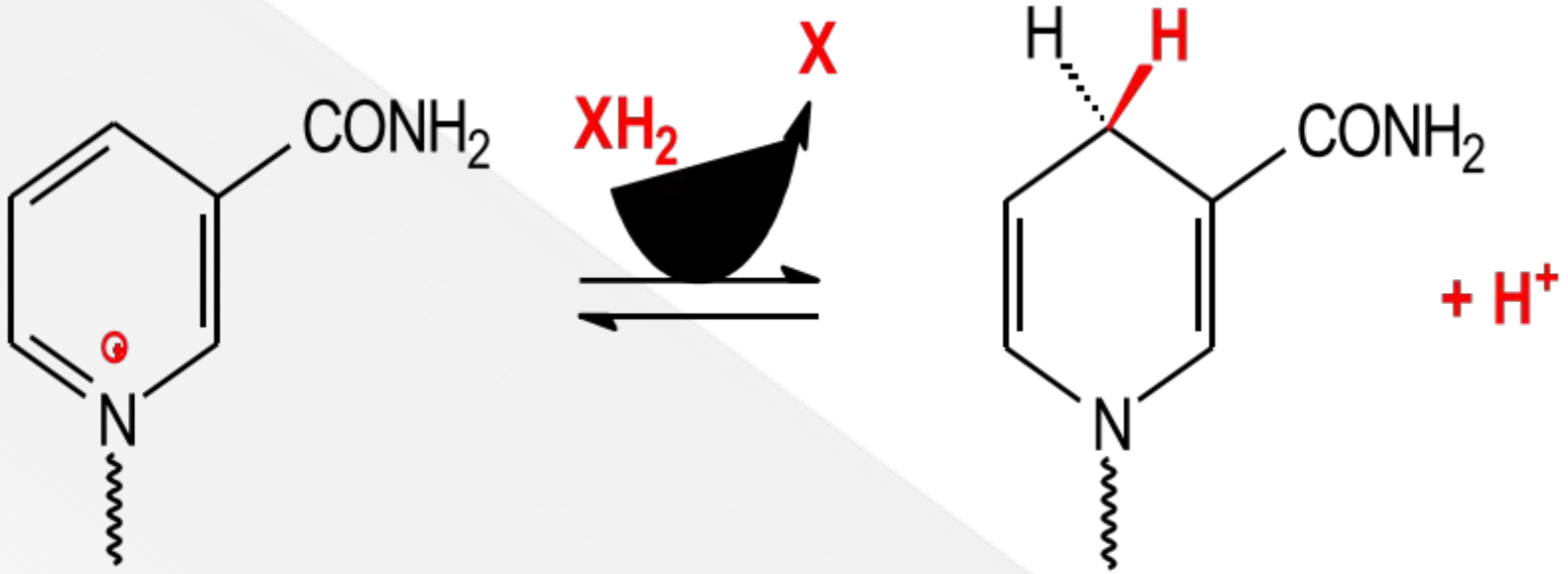


Complex I = NADH-CoQ reductase (NADH dehydro-genase complex). Complex II = Succinate-Q-reductase. Complex III = Cytochrome reductase (Cytochrome b-c1 complex). Complex IV = Cytochrome oxidase.

Complex I

- Of the two coenzymes NAD^+ & $NADP^+$, NAD^+ is more actively involved in ETC.
- Tightly bound to the inner membrane
- NAD^+ is reduced to NADH+ H^+ by dehydrogenases with the removal of two hydrogen atoms from the substrates, the substrates includes pyruvate, gly-3-P. etc.
- NADPH is more effectively utilized for anabolic reactions - fatty acid synthesis, cholesterol synthesis.

Nicotinamide coenzymes: NAD^+



oxidised coenzyme

NAD^+ or $NADP^+$

reduced coenzyme

NADH or NADPH

Flavoproteins

- The enzyme NADH dehydrogenase (NADH-coenzyme Q reductase) is a flavoprotein with FMN as the prosthetic group.
- The coenzyme FMN accepts two electrons & a proton to form $FMNH_2$

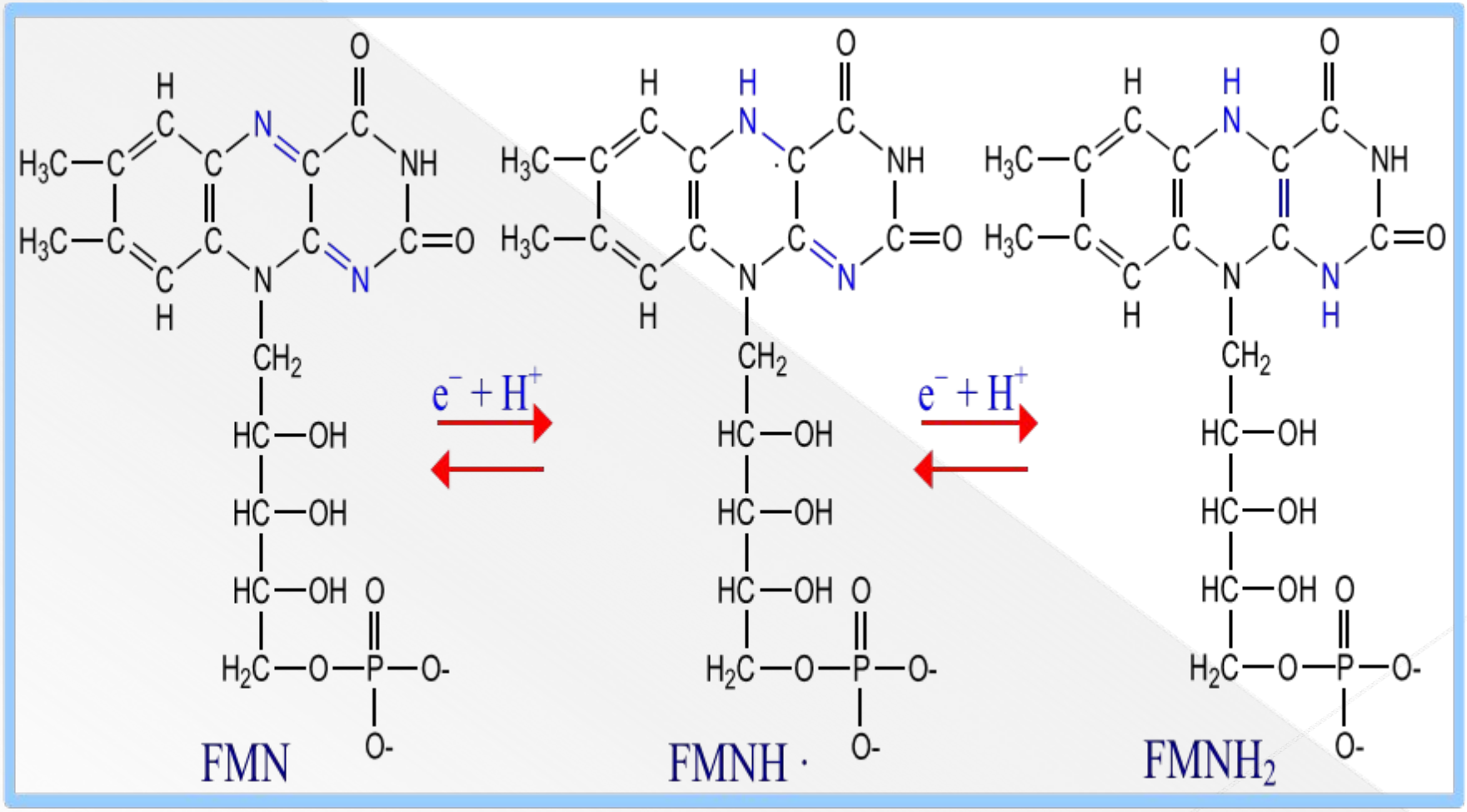
- **NADH dehydrogenase** is a complex enzyme closely associated with non-heme iron proteins or iron-sulfur proteins.
- In this, 4 protons are pumped out from mitochondria.

$$NADH + H^{+} + FMN \longrightarrow NAD + FMNH_2$$

COMPLEX II: Succinate-CoQ-Reductase

- The electrons from $FADH_2$ enter ETC at the level of Co Q.
- Succinate DH is an enzyme found in inner mitochondrial membrane.
- It is also a flavoprotein with FAD as coenzyme.
- The 3 major enzyme systems that transfer their electrons directly to ubiquinone are:
  a. Succinate dehydrogenase
  b. Fatty acyl CoA dehydrogenase
  c. Mitochondrial glycerol phosphate dehydrogenase.

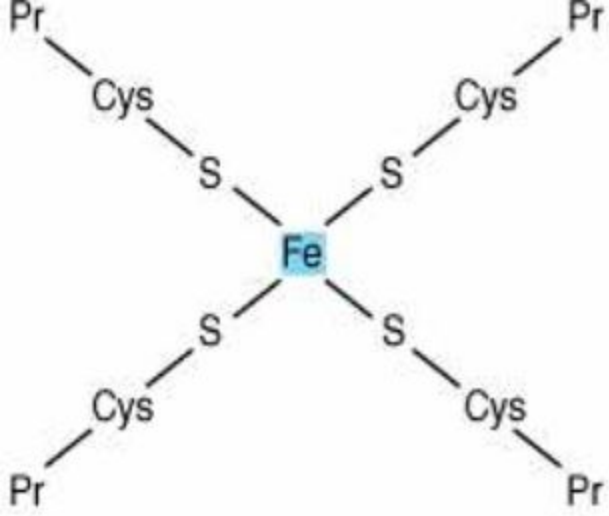
Oxidation and reduction of flavin coenzymes



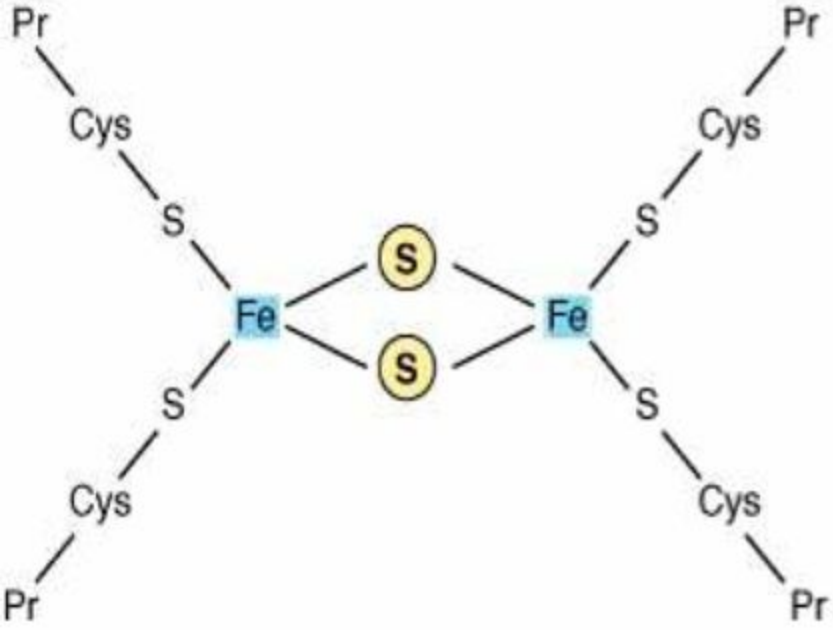
Iron-sulfur Centers (clusters)

- Iron-sulfur centers (Fe-S) are prosthetic groups containing 1-4 iron atoms
- Iron-sulfur (Fe-S) proteins exist in the oxidized (Fe^{3+}) or reduced (Fe^{2+}) state.
- Iron-sulfur centers transfer only one electron, even if they contain two or more iron atoms

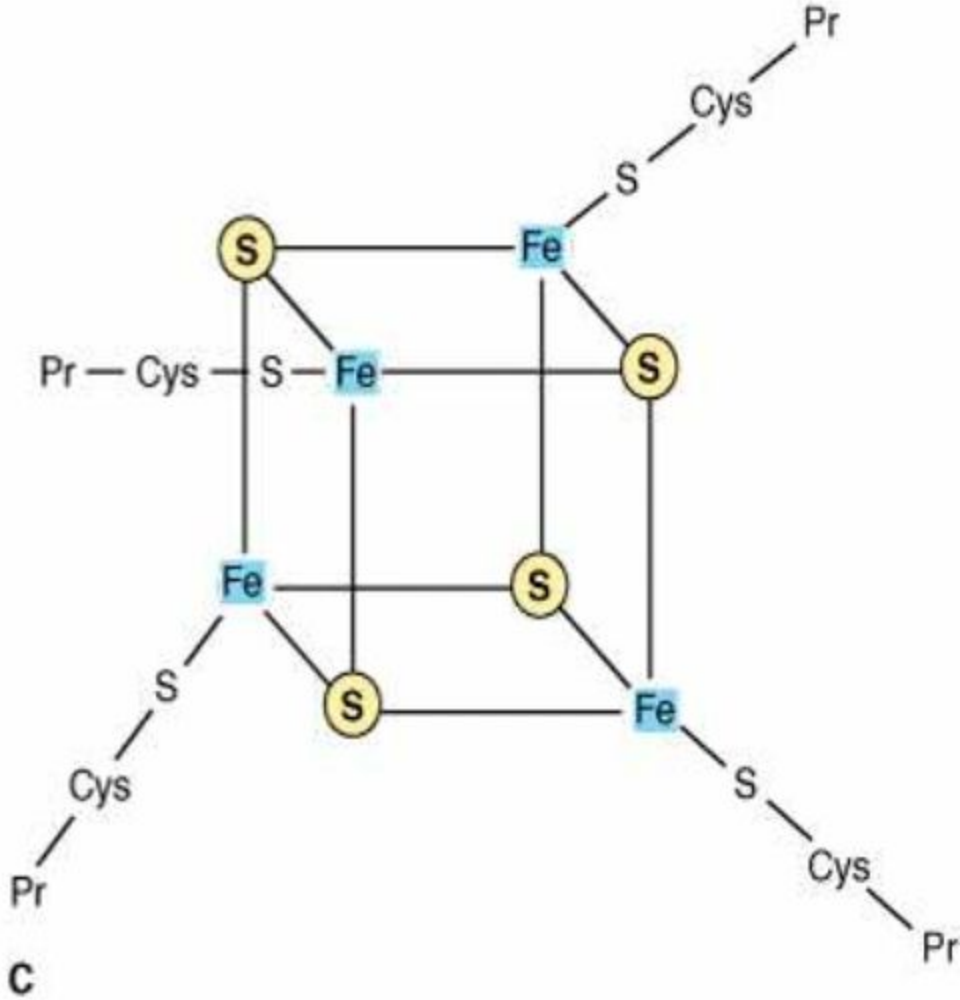
- Fe-S participates in the transfer of electrons from FMN to coenzyme Q.
- Other Fe-S proteins associated with cytochrome b & cytochrome c1 participate in the transport of electrons.


Pr
Cys
S
Fe
Pr
Cys
S
S
S
Cys
Cys
Pr
Pr

A


Pr
Cys
S
Fe
S
S
Fe
Pr
Cys
S
S
Cys
Cys
Pr
Pr

B


Pr
Cys
S
Fe
S
Pr — Cys — S — Fe
S
Fe
S
S
S
Fe
Cys
S
Pr
Cys
Pr

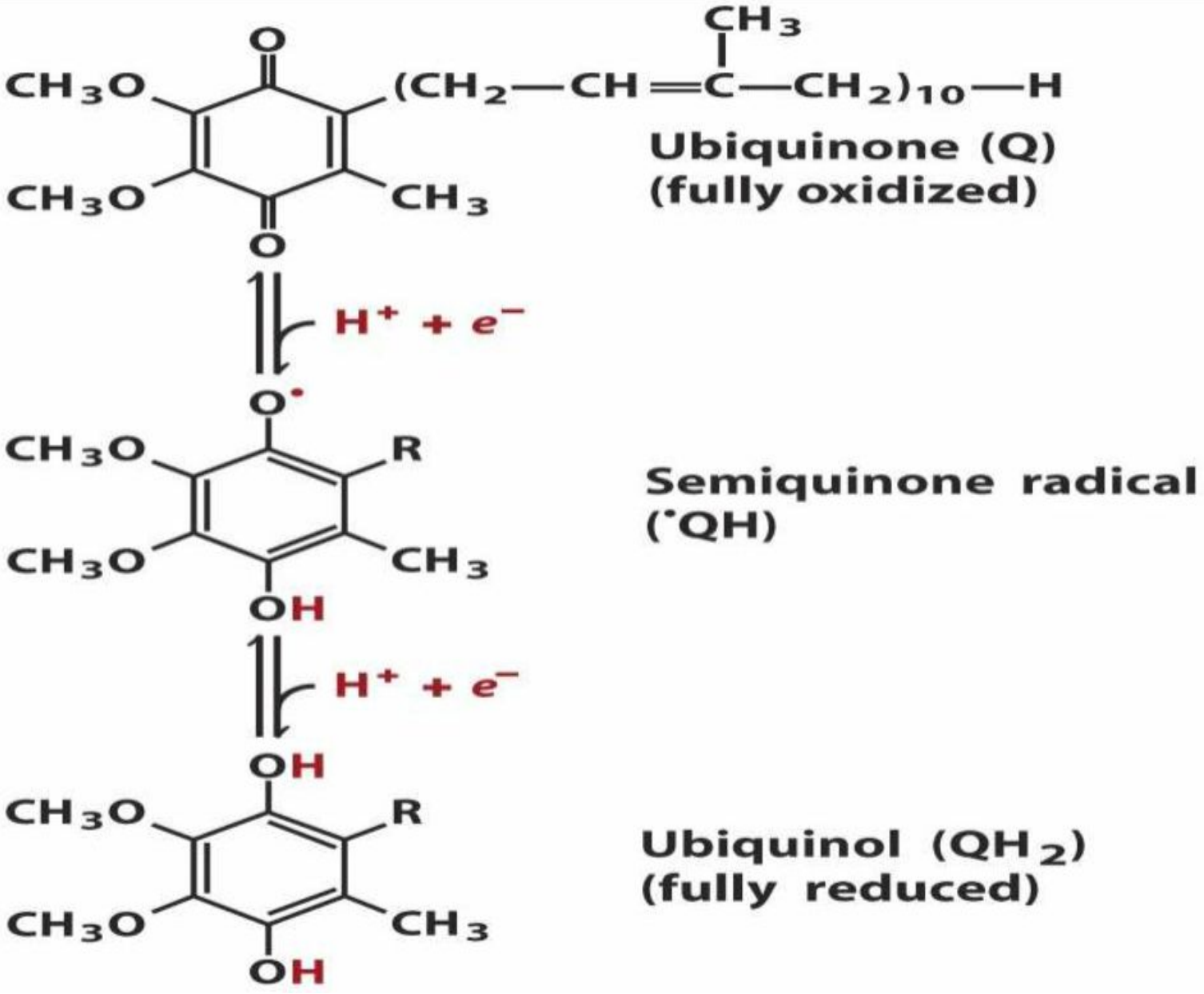
C

Coenzyme Q

- It is also known as ubiquinone.
- It is a quinone derivative with isoprenoid side chain
- Mammalian tissues possess a quinone with 10 isoprenoid units which is known as coenzyme Q_{10}
- The ubiquinone is reduced successively to semiquinone (QH) & finally to quinol (QH2)

- It accepts a pair of electrons from NADH or FADH2 through complex I or complex II respectively.
- 2 molecules of cytochrome c are reduced.
- The Q cycle facilitates the switching from the 2 electron carrier ubiquinol to the single electron carrier cytochrome c.
- This is a mobile carrier.

Coenzyme Q


CH3O
CH3O
O
O
CH3
(CH2—CH=C—CH2)10—H
CH3
Ubiquinone (Q)
(fully oxidized)
H+ + e−
O•
CH3O
CH3O
R
CH3
OH
Semiquinone radical
(•QH)
H+ + e−
OH
CH3O
CH3O
R
CH3
OH
Ubiquinol (QH2)
(fully reduced)

COMPLEX III: Cytochrome Reductase

- This is a cluster of iron-sulphur proteins, cytochrome b & cytochrome c1, both contain heme prosthetic group.
- Cytochromes are conjugated proteins
- Consists of a porphyrin ring with iron atom.
- Heme group of cytochromes differ from that found in Hb & myoglobin.

- The iron of heme in cytochromes is alternately oxidized (Fe^{3+}) & reduced (Fe^{2+})
- Which is essential for transport of electrons in the ETC.
- In this, 4 protons are pumped out.
- The electrons transported from coenzyme Q to cytochromes b, c1, c, a & a3.

- The property of reversible oxidation-reduction of heme iron present in cytochromes allows them to function as effective carriers of electrons in ETC.
- Cytochrome C:
- It is a small protein containing 104 amino acids & a heme group.
- It is a loosely bound to inner mitochondrial membrane & can be easily extracted.

COMPLEX IV: Cytochrome Oxidase (Cytochrome a & a_3)

- Contains cytochrome a and cytochrome a_3
- Which is the terminal component of ETC
- Tightly bound to inner mitochondrial membrane.
- Cytochrome oxidase is the only electron carrier, heme iron of which can directly react with molecular oxygen.

- It also contains copper that undergoes oxidation-reduction during transport of electrons.
- 2 protons are pumped out.
- In the final stage of ETC, the transported electrons, the free protons & the molecular oxygen combine to produce water.

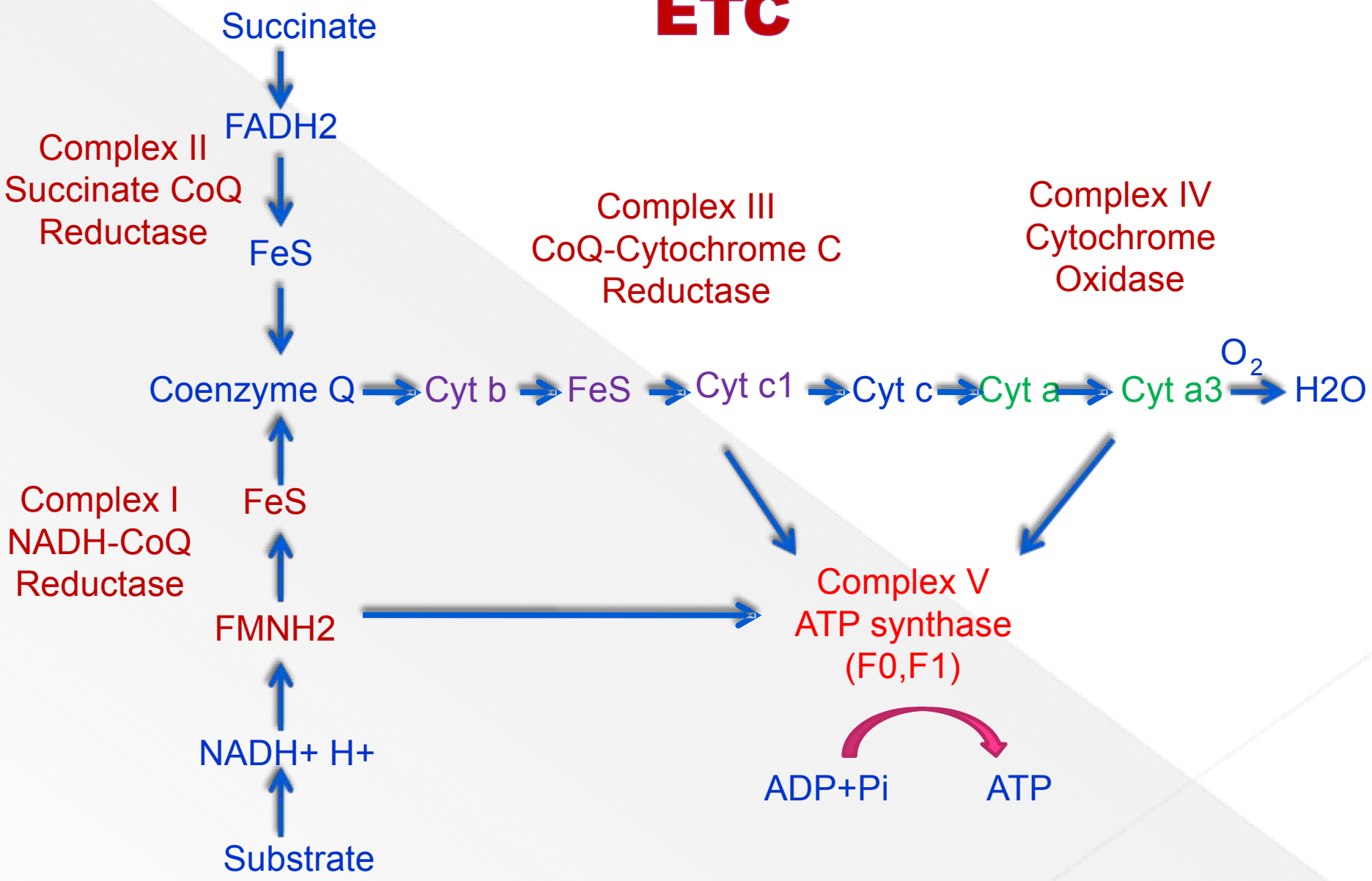
Sources of Electrons

- Electrons donors:
- NADH & $FADH_2$
- NADH: It is produced in the following reactions
- PDH complex: It transfers electrons from pyruvate to NAD^+
- α-ketoglutarate DH: It transfers electrons from alpha-ketoglutarate to NAD^+

- **Isocitrate DH:** It transfers electrons from isocitrate to NAD^+
- **Malate DH:** It transfers electrons from malate to NAD^+
- **Hydroxyacyl CoA DH:** It transfers electrons from hydroxy acyl CoA to NAD^+
- **$FADH_2$;**
- **FAD is tightly bound to enzymes called flavoproteins.**

- $FADH_2$ is produced in the following reactions.
- **Succinate DH (complex II):**
- It transfers electrons from succinate to FAD.
- **Glycerol 3-P DH:** It transfers electrons from glycerol 3-P to FAD.
- **Fatty acyl CoA DH:** It transfers electrons from fatty acids to FAD.
- $FADH_2$ donates electrons to coenzyme Q

ETC


Succinate
FADH2
Complex II
Succinate CoQ
Reductase
FeS
Complex III
CoQ-Cytochrome C
Reductase
Complex IV
Cytochrome
Oxidase
O2
Coenzyme Q
Cyt b
FeS
Cyt c1
Cyt c
Cyt a
Cyt a3
H2O
Complex I
NADH-CoQ
Reductase
FeS
Complex V
ATP synthase
(F0,F1)
FMNH2
NADH+ H+
ADP+Pi
ATP
Substrate

Inhibitors of ETC

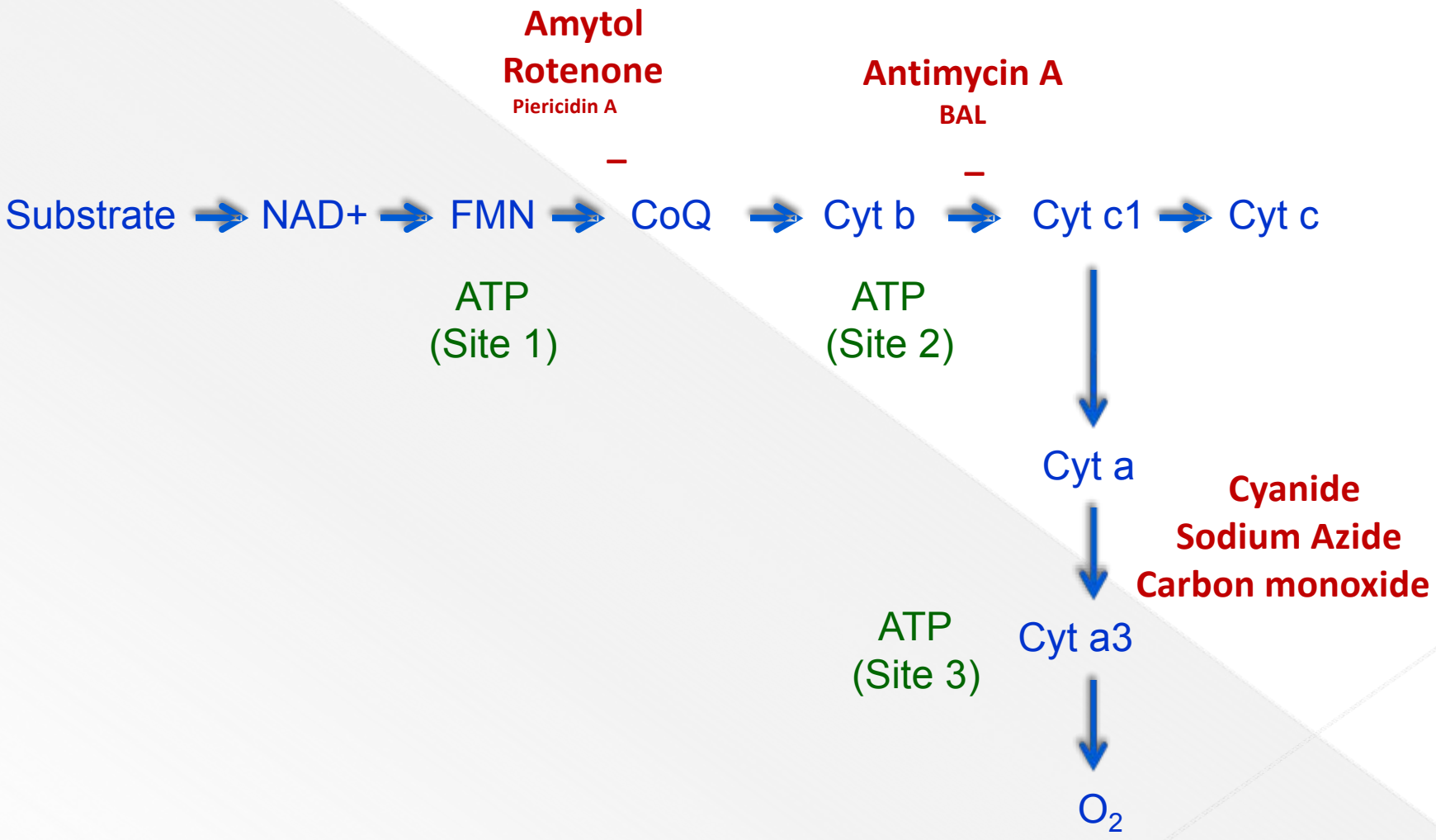
- The inhibitors bind to one of the components of ETC & block the transport of electrons
- This causes the accumulation of reduced components before the inhibitor blockade step & oxidized components after that step.
- The synthesis of ATP is dependent on ETC.

- All the site-specific inhibitors of ETC also inhibit ATP formation.
- NADH & coenzyme Q (Complex I):
- Fish poison rotenone, barbiturate drug amytol & antibiotic piercidin A inhibit this site.
- Complex II: Carboxin inhibit this site.

- Between cytochrome b & c1 (Complex III):
- Antimycin A –an antibiotic, British antilewisite (BAL) –an antidote used against war-gas-Naphthoquinone are important inhibitors of the site between cytochrome b & c1.
- Cytochrome oxidase (Complex IV):
- Carbon monoxide, cyanide, hydrogen sulphide & azide

- Effectively inhibit cytochrome while cyanide & azide react with oxidized form of cytochrome.
- Cyanide is most potent inhibitor of ETC
- It binds to Fe^{3+} of cytochrome oxidase blocking mitochondrial respiration leading to cell death.
- Cyanide poisoning causes death due to tissue asphyxia (mostly of CNS)

Sites of ATP synthesis & Inhibitors



Biological Oxidation & Oxidative Phosphorylation

- **Biological Oxidation:**
- The transfer of electrons from the reduced co-enzymes though the respiratory chain to oxygen is known as **biological oxidation.**
- Energy released during this process is **trapped as ATP.**
- This coupling of oxidation with phosphorylation is called as **OXIDATIVE PHOSPHORYLATION.**

- Complex V of the inner mitochondrial membrane is the site of oxidative phosphorylation.
- Phosphagens act as storage forms of high-energy phosphate and include creatine phosphate, which occurs in vertebrate skeletal muscle, heart, spermatozoa & brain.

- When ATP is rapidly being utilized as a source of energy for muscular contraction, phosphagens permit its concentrations to be maintained, but when the ATP/ADP ratio is high, their concentration can increase to act as a store of high-energy phosphate.

- FADH $_2$ transfers electrons to coenzyme Q thus missing the first site of oxidative phosphorylation.
- For each molecule of FADH $_2$ produces 2 ATP.

- When NADH is oxidized, about 42% of energy is trapped in the form of 3ATP & remaining is lost as heat.
- The heat liberation is not a wasteful process, since it allows ETC to go on continuously to generate ATP.
- This heat is necessary to maintain body temperature.

Mechanism of oxidative phosporylation

- Two important hypothesis to explain the process of oxidative phosphorylation.
- Namely chemical coupling & chemiosmotic
- Chemical coupling hypothesis:
- This hypothesis was put forth by Edward Slater (1953)

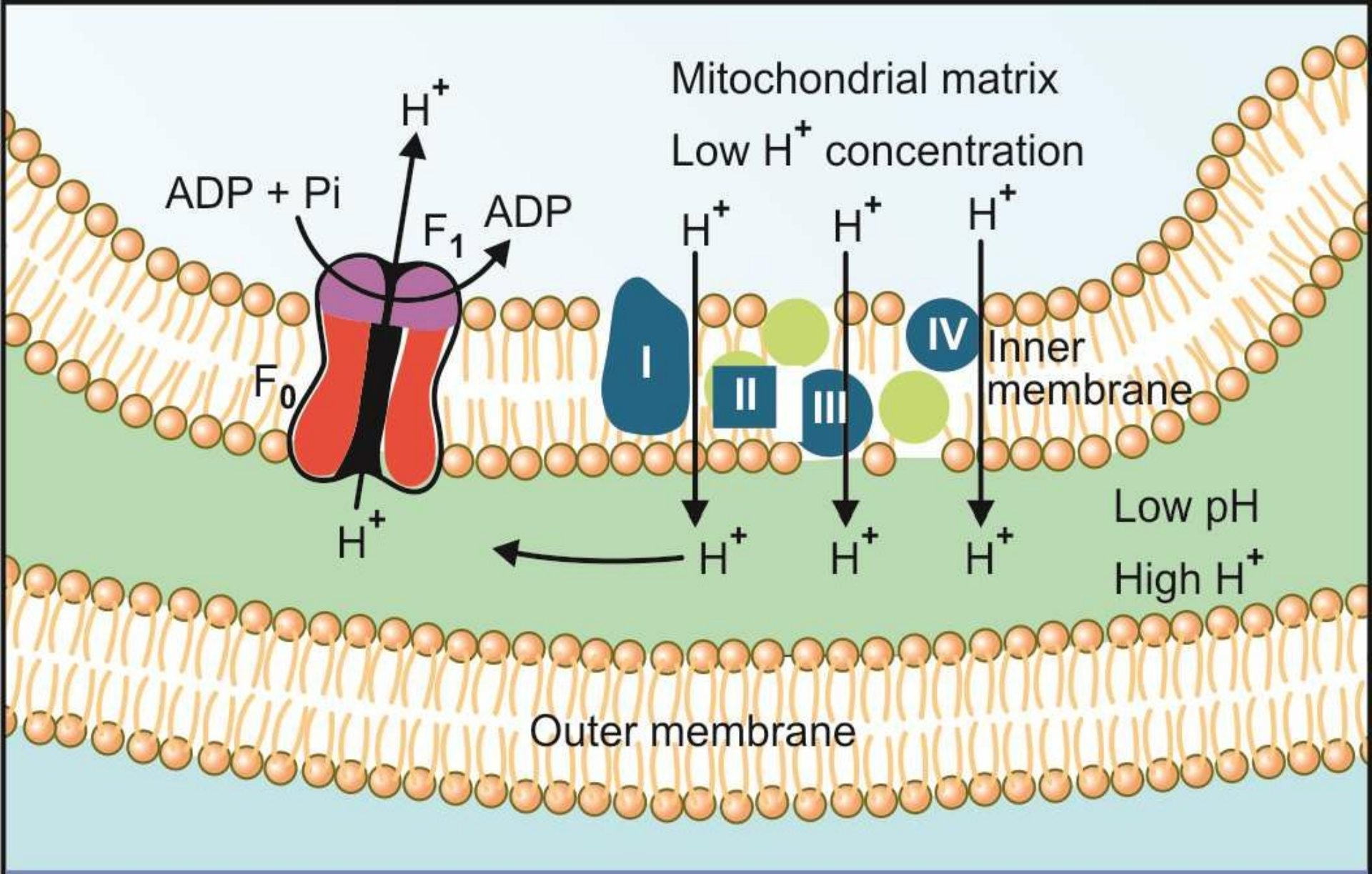
- According to this, during the course of electron transfer in respiratory chain, a series of phosphorylated high-energy intermediates are first produced which are utilized for the synthesis of ATP.
- These reactions are believed to be analogous to the substrate level phosphorylation that occurs in glycolysis or citric acid cycle.
- This hypothesis lacks experimental evidence.

Chemiosmotic Theory

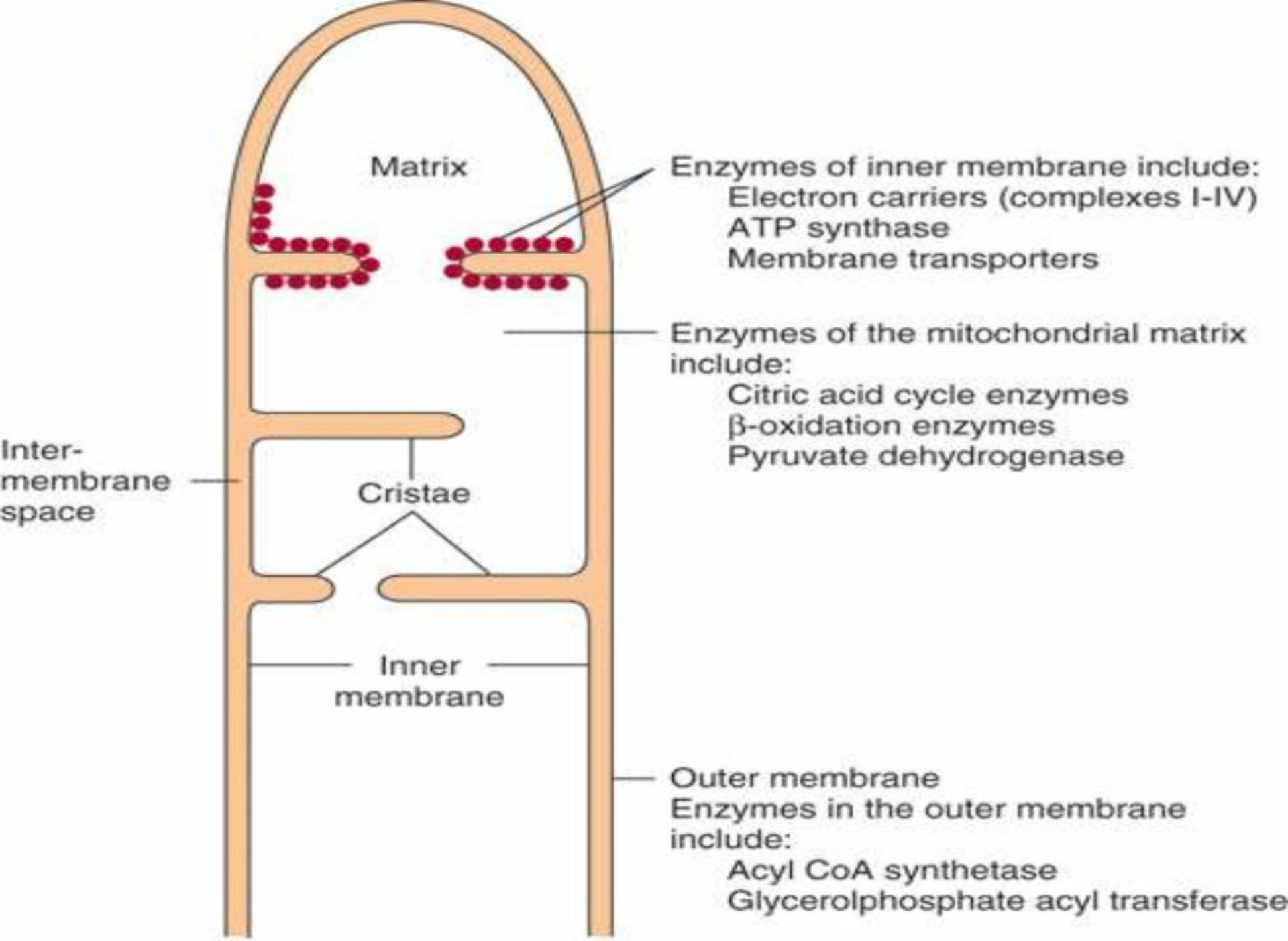
- The transport of electrons through the respiratory chain is effectively utilized to produce ATP from ADP + Pi.
- Proton gradient:
- The inner mitochondrial membrane, is impermeable to protons (H^+) & hydroxyl ions (OH).

- The transport of electrons through ETC is coupled with the translocation of protons (H^+)across the inner mitochondrial membrane from the matrix to the inter membrane space.
- The pumping of protons results in an electrochemical or proton gradient .

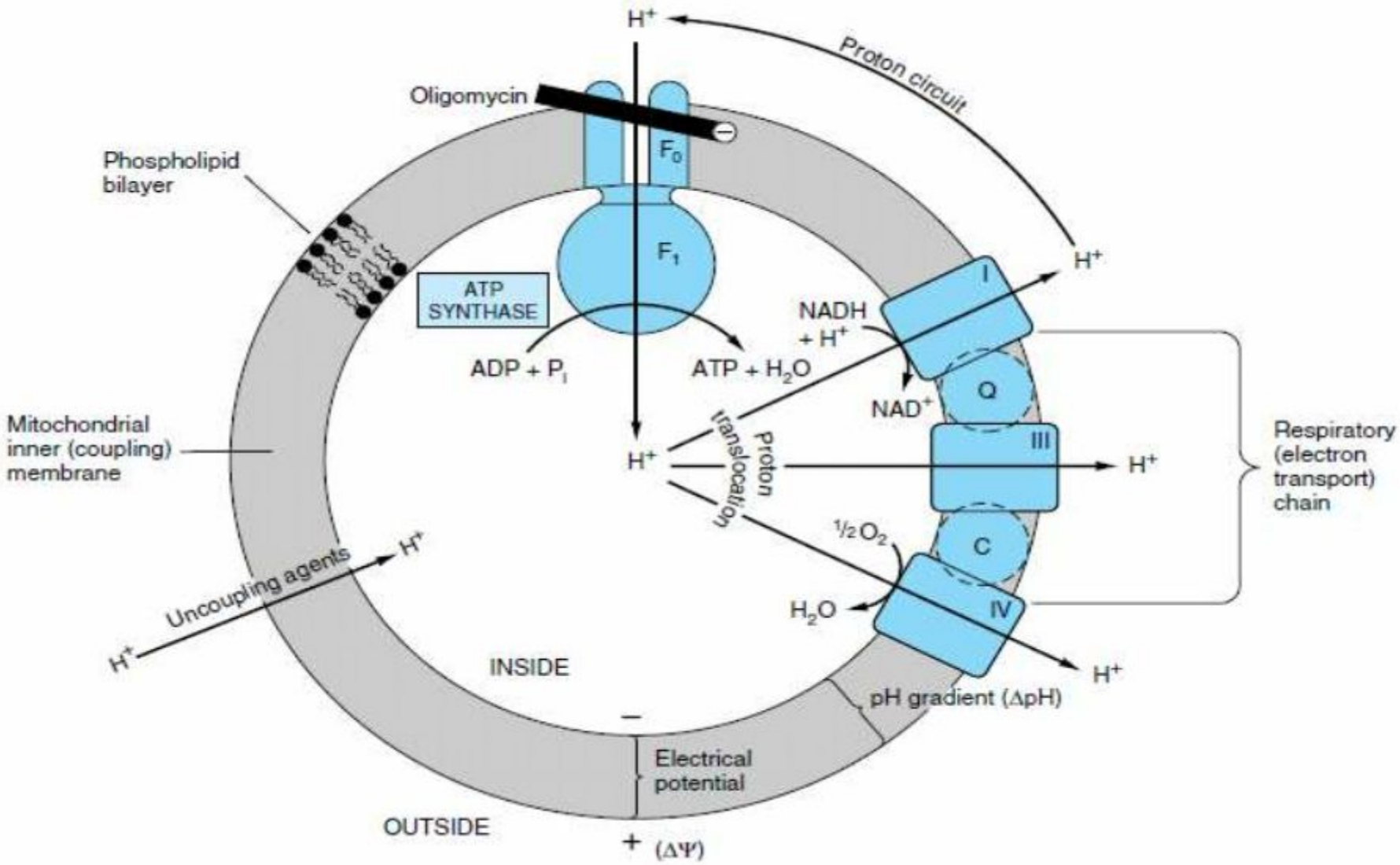
- This is due to the accumulation of more H^+ ions (low pH) on the outer side of the inner mitochondrial membrane than the inner side.
- The proton gradient developed due to the electron flow in the respiratory chain is sufficient to result in the synthesis of ATP from ADP +Pi.



I, II, III, IV = components of ETC; F0, F1, = components of ATP synthase.


Matrix
Enzymes of inner membrane include:
Electron carriers (complexes I-IV)
ATP synthase
Membrane transporters
Enzymes of the mitochondrial matrix include:
Citric acid cycle enzymes
β-oxidation enzymes
Pyruvate dehydrogenase
Inter-membrane space
Cristae
Inner membrane
Outer membrane
Enzymes in the outer membrane include:
Acyl CoA synthetase
Glycerolphosphate acyl transferase

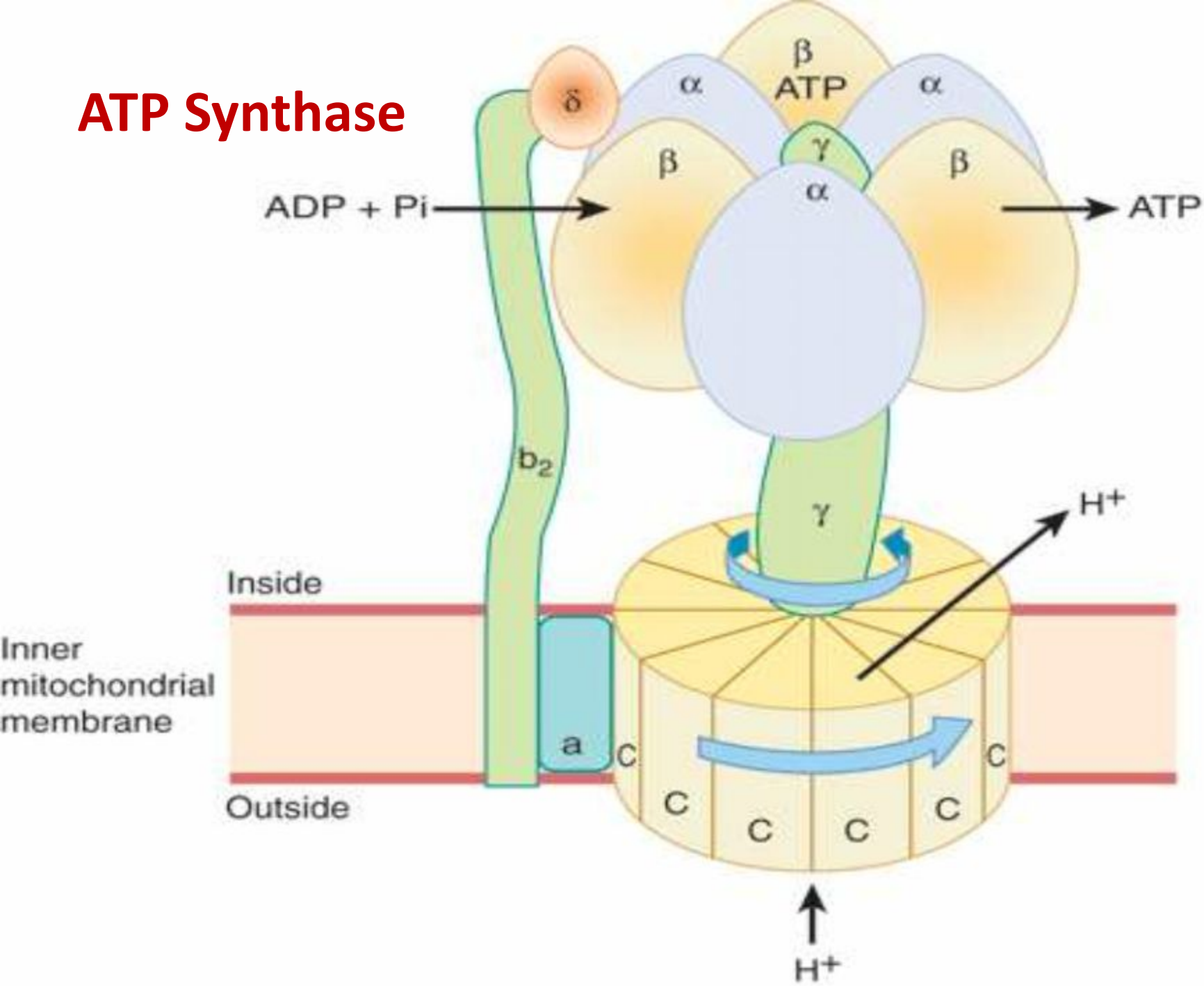
Chemiosmotic Theory



- **Enzyme systems for ATP synthesis:**
- **ATP synthase, present in the complex V, utilizes the proton gradient for the synthesis of ATP.**
- **This enzyme is also known as ATPase, since it can hydrolyze ATP to ADP + Pi.**
- **ATP synthase is a complex enzyme & consists of two functional subunits, namely F1 & Fo.**

- **Fo unit: O stands for oligomycin,**
- **Fo inhibited by oligomycin.**
- **Fo spans inner mitochondrial membrane acting as a proton channel through which protons enter the mitochondria**
- **Fo unit has 4 polypeptide chains & is connected to F1.**
- **Fo is water insoluble whereas F1 is a water soluble peripheral membrane protein.**

ATP Synthase



- The enzyme ATP synthase is Fo & F1 complex
- The Fo sub complex is composed of channel protein ‘C’ subunits to which F1-ATP synthase is attached.

Inhibitors of Oxidative phosphorylation

- The mitochondrial transport of electrons is tightly coupled with oxidative phosphorylation.
- Oxidation & phosphorylation proceed simultaneously.
- There are certain compounds that can uncouple (or delink) the electron transport from oxidative phosphorylation.

- Such compounds are known as uncouplers, increase in the permeability of inner mitochondrial membrane to protons (H^+).
- The result is that ATP synthesis does not occur
- The energy linked with the transport of electrons is dissipated as HEAT.
- The uncouplers allow (often at accelerated rate) oxidation of substrates (via NADH or $FADH_2$) without ATP formation.

- **Examples:**
- **2,4-dinitrophenol (DNP):**
- **It is small lipophilic molecule.**
- **DNP is a proton – carrier & easily diffuse through the inner mitochondrial membrane.**
- **Others –dinitrocressol, pentachlorophenol, trifluorocarbonylcyanide, phenylhydrazone**

Physiological uncouplers

- Certain physiological substances which act as uncouplers at higher concentration.
- These are thermogenin, thyroxine and long chain fatty acids & unconjugated bilirubin

- **Significance of uncoupling:**
- **The maintenance of body temperature is particularly important in hairless animals, hibernating animals & the animals adopted to cold**
- **These animals possess a specialized tissue called brown adipose tissue in the upper back & neck portions.**

- The mitochondria of brown adipose tissue are rich in electron carriers & are specialized to carry out an oxidation uncoupled from phosphorylation.
- This causes liberation of heat when fat is oxidized in the brown adipose tissue.
- The presence of brown adipose tissue in certain individuals is believed to protect them from becoming obese.

- **Thermogenin** is a natural uncoupler located in the inner mitochondrial membrane of brown adipose tissue
- It acts like an uncoupler, blocks the formation of ATP, & liberates heat.
- **Ionophores:** These are lipophilic substances that promote the transport of ions across biological membranes.
- **Valinomycin & nigercin** also act as uncouplers.

Inhibitors of oxidative phosphorylation

- **Oligomycin:** This antibiotic prevents the mitochondrial oxidation as well as phosphorylation.
- It binds with enzyme ATP synthase & blocks the proton(H^+) channels.
- Thus it prevents the translocation (re-entry) of protons into the mitochondrial matrix.

- Due to this, protons get accumulated at higher concentration in the inter membrane space
- Electron transport is stopped.
- **Atractyloside:** It is a plant toxin & inhibits oxidative phosphorylation.
- It blocks the adequate supply of ADP.

References

- Lippincott
- Harper
- Chatterjea
- Lehninger